An Iterative Network for Parameter Estimation in Nonlinear Dynamical Systems from Sparse and Partial Observations

Seong-Heon Lee*, Dongjin Lee†, Jiaxi Gu†, Joon Ha‡, and Jae-Hun Jung‡

Abstract. Estimating parameters of nonlinear dynamical systems from sparse and partial observations is a severely ill-posed inverse problem. A natural and appealing strategy is to *decouple* it: learn one network that reconstructs the unobserved (hidden) state from the observations and a current parameter guess, and another that updates the parameter from the observations and the reconstructed state, then iterate the composition to a fixed point. We study this decoupled fixed-point estimator and characterize what it can and cannot do. On the positive side, enforcing Lipschitz bounds (e.g., by spectral normalization) makes the update a contraction, so the iteration converges to a unique fixed point from any initialization at a geometric rate. On the limiting side, we show that convergence does not deliver correctness, and that the limitation is structural rather than a training artifact. Our central negative result is an incompatibility: under observational ambiguity a strict contraction cannot preserve two distinct admissible ground truths as zero-residual fixed points, so the teacher-forced target and global contractivity are mutually exclusive on a non-identifiable fiber. Because the fixed point is a deterministic function of the observation, it cannot beat the conditional-mean estimator—reconstructing the hidden state confers no information advantage at inference, since the reconstructed state is itself a function of the observation—the Bayes performance ceiling under squared loss, which a plain direct regressor typically reaches more closely. We further show, via a variational argument, that the contraction minimizing the idealized one-step risk has the conditional mean as its fixed point; and that whether that estimate is observation-consistent depends on the loss coordinate — for a curved fiber the conditional mean lies off it, but a reparameterization that straightens the fiber (logarithms for a product degeneracy) keeps it on. We diagnose the (local) identifiability of three benchmark systems a priori — SIR and Lotka–Volterra by Hermann–Krener Lie-derivative rank calculations, and OGTT by a sampled-sensitivity analysis (on the data-generating initial-condition manifold) with a scaling-symmetry ablation — and corroborate the predictions empirically. The results delineate a fundamental ceiling for deterministic decoupled point estimation and motivate lifting inference from point estimates to distributions.

Key words. dynamical systems, partial observation, inverse problems, parameter estimation, non-identifiability, machine learning, neural networks

MSC codes. 62F10, 34A55, 93B07, 68T07, 37N25

1. Introduction.

Parameter estimation for nonlinear dynamical systems from time-series observations is an inverse problem with broad scientific applications [12]. While parameter estimation is inherently ill-posed, practical constraints on data collection severely exacerbate this difficulty. In practical settings, observations are frequently restricted: the system state is only *partially* observable, capturing a limited subset of the variables, and the temporal sampling intervals can become highly *sparse*. These limitations deprive the inverse mapping of critical information, frequently driving the system into non-identifiable regimes where distinct

*Graduate School of Artificial Intelligence, POSTECH, Pohang, Republic of Korea (shlee0125@postech.ac.kr, dongjinlee@postech.ac.kr).

†Department of Mathematics, POSTECH, Pohang, Republic of Korea (jiaxigu@buffalo.edu, jung153@postech.ac.kr).

‡Department of Mathematics, Howard University, Washington, DC 20059, U.S.A. (joon.ha@Howard.edu).

parameter values induce essentially indistinguishable observed trajectories.

A common approach is to treat inference as either (i) per-instance numerical optimization over parameters (and possibly latent states) or (ii) direct regression from observations to parameters using a machine learning model. While optimization-based methods are mathematically principled, they can be computationally expensive and sensitive to initialization in sparse regimes, and they may converge to different solutions when the inverse problem admits multiple nearly equivalent explanations. Regression-based approaches amortize inference cost by learning a direct map from observations to parameters. They are simple and, as our experiments confirm, a strong baseline; what they do not do is exploit the dynamical structure of the forward model or form any explicit estimate of the unobserved state. This gap motivates methods that reconstruct the latent trajectory as part of inference, on the intuition that using the hidden state should help — an intuition whose limits this paper makes precise.

Our work sits at the intersection of several lines. Casting inference as a learned fixed-point iteration connects to deep equilibrium models [1], which likewise define an output as the fixed point of a learned operator; we use contractivity not merely for well-posedness but to expose a limitation. The recoverability of parameters is governed by *structural* versus *practical* identifiability [13], a distinction our observability analysis makes explicit for each benchmark. Finally, the remedy we point to — replacing point estimates by a posterior — is the province of simulation-based and amortized Bayesian inference [2, 9]; the conditional-mean ceiling we establish is precisely the barrier that motivates such distribution-valued inference.

In this work, we adopt a fixed-point inference formulation that makes the inference dynamics explicit. We represent sparse observations $\mathbf{x}_{\text{obs}}$ and define a learned update operator $\mathcal{T}(\theta; \mathbf{x}_{\text{obs}}) = P_\psi\left(\mathbf{x}_{\text{obs}}, H_\phi(\mathbf{x}_{\text{obs}}, \theta)\right)$, where H_ϕ is a hidden-state estimator and P_ψ is a parameter estimator. Inference is performed by the fixed-point iteration $\theta^{k+1} = \mathcal{T}(\theta^k; \mathbf{x}_{\text{obs}})$, and the final prediction is the fixed point $\theta^*(\mathbf{x}_{\text{obs}})$. The key design objective is contractivity of $\mathcal{T}(\cdot; \mathbf{x}_{\text{obs}})$ for each fixed $\mathbf{x}_{\text{obs}}$, which guarantees global convergence to a unique fixed point with a geometric rate.

Crucially, convergence does not imply correctness, and analyzing *what* the fixed point is turns out to yield a characterization of a fundamental limitation rather than a guarantee of accuracy. The core obstruction is an *incompatibility*: a strict contraction cannot keep two distinct admissible ground truths as zero-residual fixed points, since the one-step residual is bounded below by $(1 - q)$ times the distance to the operator's fixed point. Under observational ambiguity, where many parameters share an observation, the teacher-forced zero-residual target and global contractivity are therefore mutually exclusive. Because the fixed point is moreover a deterministic function of $\mathbf{x}_{\text{obs}}$, it cannot be more accurate than the conditional mean $\mathbb{E}[\theta \mid \mathbf{x}_{\text{obs}}]$ — the Bayes performance ceiling under squared loss; in particular, reconstructing the hidden state — the feature that motivates the decoupling — confers no information advantage at inference, since the reconstructed state is itself an image of $\mathbf{x}_{\text{obs}}$. The deviation of the fixed point from this ceiling is controlled by the one-step residual $\varepsilon(\mathbf{x}_{\text{obs}}) = \|\mathcal{T}(\theta^{\text{true}}; \mathbf{x}_{\text{obs}}) - \theta^{\text{true}}\|$, which we bound and decompose into a structural part and an operator-approximation part.

The consequence is sharpest under non-identifiability. Among all contractions, the one minimizing the idealized one-step risk has the conditional mean as its fixed point; and whether that estimate is observation-consistent turns out to depend on the coordinate carrying the loss — for a curved fiber the conditional mean lies *off* it (a Jensen/extreme-point effect under physical-coordinate MSE), but a reparameterization that straightens the fiber, such as logarithms for a product degeneracy, keeps it on. This regression-to-the-mean and joint-distribution collapse reflects the information content of the observations and the choice of supervised loss, not a numerical artifact; we establish it for the idealized/consistency objective and corroborate it empirically for the trained operator. Taken together, our results say that a decoupled deterministic point estimator, however well-motivated and however stably it converges, is bounded by the conditional mean in the loss coordinate — a limitation that motivates lifting inference from point estimates to distributions.

Our main contributions are:

- We formulate parameter estimation from sparse partial observations as a fixed-point problem in parameter space, defined by an update operator $\theta^{(i+1)} = \mathcal{T}(\theta^{(i)}; \mathbf{x}_{\text{obs}})$ composed of a hidden-state estimator and a parameter estimator.
- We provide a sufficient condition for $\mathcal{T}(\cdot; \mathbf{x}_{\text{obs}})$ to be contractive and hence globally convergent to a unique fixed point with a geometric rate, and we show how to enforce the required Lipschitz bounds in practice (e.g., via spectral normalization).
- We prove a contraction–ambiguity incompatibility: the one-step residual of any q-contraction is at least $(1 - q)$ times the distance to its fixed point, so two distinct admissible ground truths cannot both be zero-residual fixed points; teacher-forced correctness and global strict contractivity are structurally incompatible on a non-identifiable fiber.
- We separate convergence from correctness and characterize the fixed point: it is a function of $\mathbf{x}_{\text{obs}}$ and therefore cannot beat the Bayes squared-loss ceiling $\mathbb{E}[\theta \mid \mathbf{x}_{\text{obs}}]$ (reconstructing the hidden state adds no information at inference); its deviation from that ceiling is bounded by a one-step residual $\varepsilon(\mathbf{x}_{\text{obs}})$, which we decompose into a structural and an operator-approximation part. A variational argument shows the risk-optimal contraction has the conditional mean as its fixed point.
- We show that whether the Bayes-optimal estimate leaves the solution fiber is a property of the loss coordinate: under physical-coordinate MSE a curved fiber displaces the conditional mean off it (a Jensen/extreme-point argument), whereas a fiber-straightening reparameterization (logarithms for a product degeneracy) keeps it on — so the off-fiber phenomenon is coordinate-dependent, not intrinsic. A five-seed ablation confirms this: training the same regressor under physical- versus log-coordinate MSE moves the estimate off versus on the fiber while leaving individual-coordinate recovery unchanged.
- On epidemiological (SIR), ecological (Lotka–Volterra), and physiological (OGTT) systems, whose local identifiability we diagnose a priori — SIR and Lotka–Volterra by Hermann–Krener Lie-derivative rank, OGTT by a sampled-sensitivity analysis with a scaling-symmetry ablation — we corroborate each prediction empirically, including that a direct regressor reaches the conditional-mean ceiling more closely than the decoupled iteration, indicating the cost of the reconstruction detour.

2. Problem Formulation.

In this section, we formally define the problem of parameter estimation for dynamical systems under *partial and sparse observation* settings. Let $\mathbf{x}(t) \in \mathbb{R}^d$ denote the full state of the dynamical system and $\theta \in \Theta \subset \mathbb{R}^p$ the unknown parameter vector.

The underlying dynamics are governed by the ordinary differential equation (ODE)

$$\dot{\mathbf{x}}(t) = f(\mathbf{x}(t), \theta), \qquad t \in [0, T], \tag{2.1}$$

where $f : \mathbb{R}^d \times \Theta \to \mathbb{R}^d$ is a known, sufficiently smooth vector field. The classical parameter estimation problem seeks to infer the unknown parameter θ given the continuous trajectory of the full state $\mathbf{x}(t)$ [12]. In practical applications, however, obtaining such complete observations is rarely feasible. The following subsections define the specific constraints of realistic observations and formulates the resulting inverse problem.

2.1. Partial and Sparse Observation Setting. We consider a *partial observation* setting, in which only a subset of the state variables is observable. We decompose the full state $\mathbf{x}(t)$ into an *observed* state $x_{\text{obs}}(t) \in \mathbb{R}^{d_{\text{obs}}}$ and a *unobserved* state $x_{\text{hid}}(t) \in \mathbb{R}^{d_{\text{hid}}}$, such that $d_{\text{obs}} + d_{\text{hid}} = d$. We refer to this unobserved state as the *hidden state*. Reconstructing it explicitly is a structural choice — an inductive bias — of the decoupled estimator we study, rather than a requirement for inferring the unknown parameter θ; a direct estimator infers θ without forming it, and Section 5 examines when this reconstruction actually helps. Without loss of generality, the state vector can be partitioned as $\mathbf{x}(t) = \left[x_{\text{obs}}(t)^\top, x_{\text{hid}}(t)^\top\right]^\top$. To formalize this limitation, we define a non-injective map $h : \mathbb{R}^d \to \mathbb{R}^{d_{obs}}$ acting as a coordinate projection:

$$x_{\text{obs}}(t) = h(\mathbf{x}(t)). \tag{2.2}$$

The data are further constrained by *sparse sampling* at discrete times. We assume a uniform sampling scheme over the time interval $[0, T]$ with a fixed step size Δt:

$$0 = t_0 < t_1 < \cdots < t_N = T, \tag{2.3}$$

where $N\Delta t = T$. The term *sparse* specifically denotes scenarios where $(N+1)$ is extremely small relative to the intrinsic time scale of the dynamics. Consequently, the partial and sparse time-series data collected from the system is denoted as the set of discrete observations $\{x_{\text{obs}}(t_i)\}_{i=0}^N$.

Since our estimators are modeled as multi-layer perceptrons (MLPs), the discrete sampled trajectories must be flattened into 1D vectors. We write the flattened observation vector $\mathbf{x}_{\text{obs}} \in \mathbb{R}^{(N+1)\times d_{\text{obs}}}$ and the hidden state vector $\mathbf{x}_{\text{hid}} \in \mathbb{R}^{(N+1)\times d_{\text{hid}}}$ by concatenating the state vectors at each sampling time:

$$\mathbf{x}_{\text{obs}} := [x_{\text{obs}}(t_0)^\top, \ldots, x_{\text{obs}}(t_N)^\top]^\top, \tag{2.4}$$
$$\mathbf{x}_{\text{hid}} := [x_{\text{hid}}(t_0)^\top, \ldots, x_{\text{hid}}(t_N)^\top]^\top. \tag{2.5}$$

Estimating the unknown parameter vector θ from the partial and sparse observations $\mathbf{x}_{\text{obs}}$ constitutes an inverse problem involving unobserved state trajectories. Since the hidden state

$\mathbf{x}_{\text{hid}}$ is unmeasured yet continuously coupled with the observed dynamics through the governing ODE, traditional joint optimization over both states and parameters frequently leads to highly non-convex and ill-conditioned objective landscapes. This computational difficulty is significantly amplified in sparse sampling regimes, where classical solvers are highly sensitive to initialization and prone to getting trapped in local minima or diverging. To alleviate these numerical challenges, a structured inference framework that systematically decouples the state reconstruction and parameter estimation tasks is required.

Remark 2.1 (Why sparsity is essential). Given the *continuous* partial observation $x_{\text{obs}}(t)$ for all $t \in [0, T]$, classical theory offers recovery guarantees: Takens' embedding theorem [11] and the Hermann–Krener observability rank condition [5] (Section 6) certify, under genericity assumptions, that the full state and the parameters can in principle be reconstructed from the observed component and its time derivatives. These guarantees rely on access to high-order temporal information (derivatives, or equivalently a dense time series). Under the sparse sampling of (2.3), that information is unavailable, since derivatives cannot be reliably estimated from a handful of points. The guarantees then no longer apply, and the problem may become practically, or even structurally, non-identifiable. This is the regime we study, and it is what makes the inverse problem genuinely hard.

3. Method: Iteration of Two Networks for Parameter Estimation.

This section presents an iterative neural network framework for parameter estimation from partial and sparse observations. To motivate our approach, consider the forward mapping from the parameter to the partial observations. As illustrated in Figure 1, this mapping consists of two distinct steps: integrating the governing ODE $\dot{\mathbf{x}} = f(\mathbf{x}; \theta)$ to obtain the full state $\mathbf{x} = [\mathbf{x}_{\text{obs}}, \mathbf{x}_{\text{hid}}]^\top$, and applying a non-injective projection $h(\mathbf{x}) = \mathbf{x}_{\text{obs}}$ to obtain the partial observation.

A direct regression from partial observations to parameters bypasses this two-step structure and never forms an explicit estimate of the hidden state $\mathbf{x}_{\text{hid}}$. Because different hidden states and parameters can generate the same partial observation, it is tempting to conclude that ignoring $\mathbf{x}_{\text{hid}}$ must discard information and that a method reconstructing it should do better. This intuition motivates the decoupled design developed below; we return to it critically in Section 5, where we show that at inference the reconstructed hidden state is itself a function of the observation and therefore confers no such information advantage.

Instead of a direct map, we invert the forward mapping step-by-step. We decompose the inverse problem into two stages: (i) inferring the full state $\mathbf{x}$ from the partial observation $\mathbf{x}_{\text{obs}}$, and (ii) estimating the parameter θ from the full state $\mathbf{x} = [\mathbf{x}_{\text{obs}}, \mathbf{x}_{\text{hid}}]^\top$. However, this decomposition introduces a circular dependency. Estimating the full state $\mathbf{x}$ requires the parameter θ to resolve the non-injectivity of the projection h, and estimating the parameter θ requires the full state $\mathbf{x}$.

To resolve this dependency, we propose the iterative neural network framework shown in Figure 2. We introduce two neural networks designed to mimic the step-by-step inversion by acting as conditional estimators. The hidden-state estimator $H_\phi : (\mathbf{x}_{\text{obs}}, \theta) \mapsto \mathbf{x}_{\text{hid}}$ estimates the hidden state $\mathbf{x}_{\text{hid}}$ from the observation $\mathbf{x}_{\text{obs}}$ conditioned on a specific parameter guess θ. Subsequently, the parameter estimator $P_\psi : (\mathbf{x}_{\text{obs}}, \mathbf{x}_{\text{hid}}) \mapsto \theta$ refines the parameter guess using the completed trajectory. These networks are trained independently using decoupled supervised learning to prevent error accumulation, as illustrated in Figure 2a. During infer-

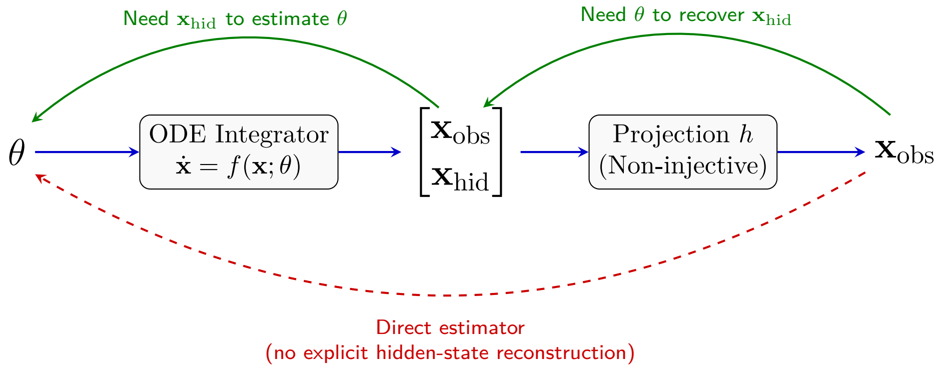


Figure 1: Illustration of the forward mapping and the inherent circular dependency in the inverse problem. The forward process generates the full state via an ODE integrator, followed by a non-injective projection h that yields the partial observation $\mathbf{x}_{\text{obs}}$. A direct estimator (red dashed arrow) maps the observation to the parameter without forming the hidden state. Inverting this process step-by-step requires resolving a circular dependency: estimating the parameter θ requires the hidden state $\mathbf{x}_{\text{hid}}$ (left green arrow), while recovering $\mathbf{x}_{\text{hid}}$ necessitates prior knowledge of θ (right green arrow).

ence, they are coupled to form a fixed-point iteration operator, continuously updating the parameter estimate as follows:

$$\hat{\theta}^{(k+1)} = \mathcal{T}(\hat{\theta}^{(k)}; \mathbf{x}_{\text{obs}}) := P_\psi(\mathbf{x}_{\text{obs}}, H_\phi(\mathbf{x}_{\text{obs}}, \hat{\theta}^{(k)})). \tag{3.1}$$

By alternating between the hidden state and parameter updates, the estimate converges to a network-consistent fixed point θ^* that resolves the circular dependency (whether this network-consistent point is also observation-consistent is the subject of Section 5). The remainder of this section details the data representation, the iteration operator $\mathcal{T}$, and the training and inference algorithms.

3.1. Data Representation. To train the estimators via supervised learning, we generate a synthetic dataset by simulating the dynamics. We sample parameters θ and initial conditions $\mathbf{x}(t_0)$ from a predefined training distribution $\mathcal{P}_{\text{train}}$, solve the initial value problem for the ODE (2.1) on $[0, T]$ via a numerical solver, and record the states at the sampling times $t_0, \ldots, t_N$. This procedure yields an i.i.d. dataset $\mathcal{D}$ of M trajectories:

$$\mathcal{D} := \left\{ (\mathbf{x}_{\text{obs}}^{(m)}, \mathbf{x}_{\text{hid}}^{(m)}, \theta^{(m)}) \right\}_{m=1}^{M}, \tag{3.2}$$

where $\mathbf{x}_{\text{obs}}^{(m)}$ and $\mathbf{x}_{\text{hid}}^{(m)}$ are the flattened vectors defined in Section 2. Prior to training, these vectors are scaled via a dataset-level normalization scheme (fit on the training split only) to ensure numerical stability.

3.2. Fixed-Point Iteration Operator. Given an observation $\mathbf{x}_{\text{obs}}$, our objective is to accurately infer the corresponding system parameter θ in accordance with the data generation process as illustrated in Figure 1. We formulate this parameter estimation as finding a fixed-point of a discrete dynamical system defined by an iterative operator. To achieve this, we decompose the estimation problem into two subtasks and construct mappings to perform each.

The first mapping is the hidden-state estimator H_ϕ, which solves a full state recovery task:

$$H_\phi : \mathbb{R}^{(N+1)d_{\text{obs}}} \times \mathbb{R}^p \to \mathbb{R}^{(N+1)d_{\text{hid}}}, \quad (\mathbf{x}_{\text{obs}}, \theta) \mapsto \hat{\mathbf{x}}_{\text{hid}}. \tag{3.3}$$

This mapping takes the observation $\mathbf{x}_{\text{obs}}$ and a specific parameter guess θ as inputs to estimate the corresponding hidden state $\hat{\mathbf{x}}_{\text{hid}}$. Conditioning on θ supplies H_ϕ with side information that the projection h discards, so that it can estimate a full state trajectory $\mathbf{x} = [\mathbf{x}_{\text{obs}}, \hat{\mathbf{x}}_{\text{hid}}]^\top$ consistent with the underlying dynamics; we emphasize that this addresses the non-injectivity of h only relative to the supplied guess, and does not by itself resolve the non-identifiability of θ from $\mathbf{x}_{\text{obs}}$ (Section 5).

The second mapping is the parameter estimator P_ψ, which solves a parameter estimation task from the full-state:

$$P_\psi : \mathbb{R}^{(N+1)d_{\text{obs}}} \times \mathbb{R}^{(N+1)d_{\text{hid}}} \to \mathbb{R}^p, \quad (\mathbf{x}_{\text{obs}}, \mathbf{x}_{\text{hid}}) \mapsto \hat{\theta}. \tag{3.4}$$

This mapping deduces the parameter vector from the reconstructed full-state trajectory.

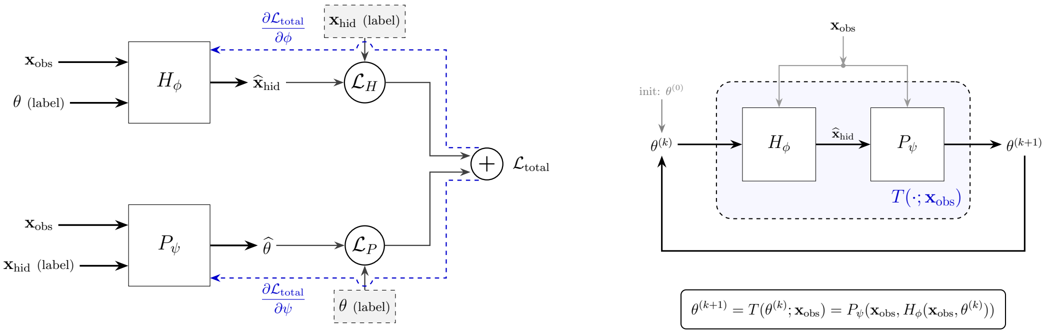
We implement both mappings as multi-layer perceptrons (MLPs) parameterized by ϕ and ψ, respectively; the choice of a flexible, expressive hypothesis class is what makes each conditional mapping learnable from data.

By composing these two networks, we define the iteration operator $\mathcal{T}(\cdot; \mathbf{x}_{\text{obs}}) : \Theta \to \Theta$, given by

$$\mathcal{T}(\theta; \mathbf{x}_{\text{obs}}) := P_\psi(\mathbf{x}_{\text{obs}}, H_\phi(\mathbf{x}_{\text{obs}}, \theta)). \tag{3.5}$$

This composite operator acts as a discrete dynamical system in parameter space. We refer to $\mathcal{T}$ as the *fixed-point iteration operator*. Finding a fixed point θ^* such that $\theta^* = \mathcal{T}(\theta^*; \mathbf{x}_{\text{obs}})$ provides a mechanism to resolve the cyclic dependency between two subtasks.

3.3. Decoupled Supervised Training. We train the networks H_ϕ and P_ψ in a supervised manner using the synthetic dataset $\mathcal{D}$ generated in Section 3.1. As illustrated in Figure 2a, we train the two networks independently without cyclic dependency to prevent error accumulation. This decoupled approach is conceptually analogous to *teacher forcing* [15]. During inference, H_ϕ must rely on a previously estimated parameter $\hat{\theta}^{(k)}$ as its conditional input, and P_ψ must rely on the estimated hidden state $\hat{\mathbf{x}}_{\text{hid}}^{(k+1)}$. If trained jointly in this coupled manner, early estimation errors would propagate and amplify across the networks. In our teacher-forced training, we break this cycle by substituting the unavailable inference-phase estimates with exact ground-truth values. Specifically, we provide the true parameter θ^{true} as the conditional input to H_ϕ (targeting the true $\mathbf{x}_{\text{hid}}^{\text{true}}$), and provide the true state $\mathbf{x}_{\text{hid}}^{\text{true}}$ as the conditional input to P_ψ (targeting the true θ^{true}).



(a) Supervised training via teacher forcing

(b) Inference via alternating iteration

Figure 2: Overview of the proposed fixed-point iteration networks. The training and inference pipelines are connected but structurally distinct. **(a)** During training, the hidden-state estimator H_ϕ and the parameter estimator P_ψ are decoupled. Ground-truth values are provided to optimize $\mathcal{L}_H$ and $\mathcal{L}_P$ independently. **(b)** At inference, given sparse observations $\mathbf{x}_{\text{obs}}$, the networks are coupled to define a fixed-point update operator $\mathcal{T}(\cdot; \mathbf{x}_{\text{obs}})$.

Given a training sample $(\mathbf{x}_{\text{obs}}, \mathbf{x}_{\text{hid}}^{\text{true}}, \theta^{\text{true}}) \sim \mathcal{D}$, we optimize the network parameters ϕ and ψ by minimizing their respective mean squared error (MSE) loss functions:

$$\mathcal{L}_H(\phi) := \mathbb{E}_{\mathcal{D}} \left[\left\| H_\phi(\mathbf{x}_{\text{obs}}, \theta^{\text{true}}) - \mathbf{x}_{\text{hid}}^{\text{true}} \right\|_2^2 \right], \tag{3.6}$$

$$\mathcal{L}_P(\psi) := \mathbb{E}_{\mathcal{D}} \left[\left\| P_\psi(\mathbf{x}_{\text{obs}}, \mathbf{x}_{\text{hid}}^{\text{true}}) - \theta^{\text{true}} \right\|_2^2 \right]. \tag{3.7}$$

Because the two networks do not feed their outputs to one another during training, their computational graphs are strictly disconnected. Consequently, when optimizing the total loss $\mathcal{L}_{\text{total}} := \mathcal{L}_H(\phi) + \mathcal{L}_P(\psi)$ via stochastic gradient descent (SGD), the parameter updates are completely decoupled. As depicted in Figure 2a, the gradients of the total loss reduce to the partial derivatives with respect to each network's parameters:

$$\frac{\partial \mathcal{L}_{\text{total}}}{\partial \phi} = \frac{\partial \mathcal{L}_H}{\partial \phi}, \qquad \frac{\partial \mathcal{L}_{\text{total}}}{\partial \psi} = \frac{\partial \mathcal{L}_P}{\partial \psi}. \tag{3.8}$$

These gradients are computed and applied independently without backpropagating errors across the networks.

This teacher-forced protocol *targets* an operator that preserves the ground truth,

$$\mathcal{T}(\theta^{\text{true}}; \mathbf{x}_{\text{obs}}) \approx \theta^{\text{true}}.$$

Whether the resulting fixed point actually coincides with the ground truth is a separate question, which we take up in Section 5: there we show that this target is in general unreachable under observational ambiguity, that the fixed point is bounded by the conditional mean, and that under non-identifiability the conditional mean is displaced off the solution fiber.

Algorithm 3.1 Fixed-step iterative inference for parameter estimation

Require: observed samples $\mathbf{x}_{\text{obs}} = [x_{\text{obs}}(t_0), \dots, x_{\text{obs}}(t_N)]^\top$ ($N+1$ times); trained networks H_ϕ, P_ψ; number of iterations K

1: initialize $\hat{\theta}^{(0)}$ (e.g., centered at the dataset mean)
2: **for** $k = 0, 1, \dots, K-1$ **do**
3: $\quad \widehat{\mathbf{x}}_{\text{hid}}^{(k+1)} \leftarrow H_\phi(\mathbf{x}_{\text{obs}}, \widehat{\theta}^{(k)})$
4: $\quad \widehat{\theta}^{(k+1)} \leftarrow P_\psi(\mathbf{x}_{\text{obs}}, \widehat{\mathbf{x}}_{\text{hid}}^{(k+1)})$
5: **end for**
6: **return** $\widehat{\theta}^{(K)}$

3.4. Coupled Inference via Fixed-Point Iteration.

Although training is strictly decoupled, inference requires coupling the two networks to form the fixed-point iteration operator $\mathcal{T}$, as illustrated in Figure 2b. At inference time, only the sparse and partial observation $\mathbf{x}_{\text{obs}}$ is available; the ground-truth parameter and hidden states are completely unknown.

Starting from an initial parameter guess $\hat{\theta}^{(0)}$, we sequentially apply the alternating updates between the hidden-state estimator and the parameter estimator. The overall inference procedure is summarized in Algorithm 3.1.

The iterative process defined in Algorithm 3.1 is mathematically equivalent to computing the orbit of the discrete dynamical system $\hat{\theta}^{(k+1)} = \mathcal{T}(\hat{\theta}^{(k)}; \mathbf{x}_{\text{obs}})$. For this iteration to be stable and independent of the initial guess, the operator $\mathcal{T}(\cdot; \mathbf{x}_{\text{obs}})$ must be a contraction mapping. We establish sufficient conditions for this contractivity in Section 4, showing that enforcing appropriate Lipschitz constraints on H_ϕ and P_ψ guarantees global geometric convergence to a unique fixed point $\theta^*(\mathbf{x}_{\text{obs}})$. Because of this exponential convergence rate, a very small number of iterations (e.g., $K = 1$ to 10) typically suffices in practice to reach the fixed point.

4. Theory 1: Convergence.

In this section we analyze the iteration operator $\theta^{(k+1)} = \mathcal{T}(\theta^{(k)}; \mathbf{x}_{\text{obs}})$ introduced in 3. The introduction of the iteration raises two critical theoretical questions:

Convergence: Does this iteration process stably converge to a unique fixed point? Sensitivity: Is it sensitive to the initial guess $\theta^{(0)}$?

In this section, we prove that by enforcing strict Lipschitz bounds on two neural networks H_ϕ and P_ψ, the composite operator $\mathcal{T}$ becomes a global contraction mapping, guaranteeing exponential convergence to a unique fixed point θ^* from any arbitrary initialization.

4.1. Preliminaries: contraction mappings and Banach's theorem.

Let $(\Theta, \|\cdot\|_2)$ be a complete metric space. [1] A map $\mathcal{T} : \Theta \to \Theta$ is a *contraction* if there exists $K \in [0, 1)$ such that

$$\|\mathcal{T}(\theta_a) - \mathcal{T}(\theta_b)\|_2 \le K\|\theta_a - \theta_b\|_2, \qquad \forall\, \theta_a, \theta_b \in \Theta. \tag{4.1}$$

By *Banach's fixed-point theorem*, a contraction has a unique fixed point θ^* and the iteration $\theta^{(k+1)} = \mathcal{T}(\theta^{(k)})$ converges to θ^* for any initialization. Moreover, the convergence is geometric:

[1] In practice, Θ is taken as a closed subset of $\mathbb{R}^p$ in normalized coordinates.

$$\|\theta^{(k)} - \theta^*\|_2 \leq K^k \|\theta^{(0)} - \theta^*\|_2. \tag{4.2}$$

4.2. A Sufficient Condition for Contractivity of $\mathcal{T}$. Recall that our update operator is the composition:

$$\mathcal{T}(\theta; \mathbf{x}_{\text{obs}}) = P_\psi\big(\mathbf{x}_{\text{obs}}, H_\phi(\mathbf{x}_{\text{obs}}, \theta)\big). \tag{4.3}$$

We provide a simple sufficient condition for (4.1) in terms of Lipschitz constants of the two networks.

Assumption 4.1 (Lipschitz continuity of the learned maps). For each fixed input $\mathbf{x}_{\text{obs}}$, the hidden-state estimator H_ϕ is Lipschitz in θ: there exists $\text{Lip}_H \geq 0$ such that

$$\|H_\phi(\mathbf{x}_{\text{obs}}, \theta_a) - H_\phi(\mathbf{x}_{\text{obs}}, \theta_b)\|_2 \leq \text{Lip}_H \|\theta_a - \theta_b\|_2, \qquad \forall\, \theta_a, \theta_b \in \Theta.$$

Moreover, the parameter estimator P_ψ is Lipschitz in its hidden-state input: there exists $\text{Lip}_P \geq 0$ such that for all $\mathbf{z}_a, \mathbf{z}_b$ in the range of $H_\phi(\mathbf{x}_{\text{obs}}, \cdot)$,

$$\|P_\psi(\mathbf{x}_{\text{obs}}, \mathbf{z}_a) - P_\psi(\mathbf{x}_{\text{obs}}, \mathbf{z}_b)\|_2 \leq \text{Lip}_P \|\mathbf{z}_a - \mathbf{z}_b\|_2.$$

Theorem 4.2 (Contractivity of the Iterative Operator). *Under Assumption 4.1, the update operator $\mathcal{T}(\cdot; \mathbf{x}_{\text{obs}})$ is Lipschitz with constant at most $\text{Lip}_{\mathcal{T}} := \text{Lip}_P \text{Lip}_H$:*

$$\|T(\theta_a; \mathbf{x}_{\text{obs}}) - T(\theta_b; \mathbf{x}_{\text{obs}})\|_2 \leq \text{Lip}_P \text{Lip}_H \|\theta_a - \theta_b\|_2, \qquad \forall\, \theta_a, \theta_b \in \Theta.$$

In particular, if $\text{Lip}_P \text{Lip}_H < 1$, then $\mathcal{T}(\cdot; \mathbf{x}_{\text{obs}})$ is a contraction on $(\Theta, \|\cdot\|_2)$ and hence has a unique fixed point $\theta^*(\mathbf{x}_{\text{obs}})$. Moreover, for any initialization $\theta^{(0)} \in \Theta$, the iteration $\theta^{(k+1)} = \mathcal{T}(\theta^{(k)}; \mathbf{x}_{\text{obs}})$ converges to $\theta^*(\mathbf{x}_{\text{obs}})$ and satisfies the geometric rate bound (4.2) with $\text{Lip}_{\mathcal{T}}$.

Proof. Fix $\mathbf{x}_{\text{obs}}$ and let $\theta_a, \theta_b \in \Theta$. By (4.3) and the Lipschitz property of P_ψ in its second argument,

$$\begin{aligned}\|\mathcal{T}(\theta_a; \mathbf{x}_{\text{obs}}) - \mathcal{T}(\theta_b; \mathbf{x}_{\text{obs}})\|_2 &= \|P_\psi(\mathbf{x}_{\text{obs}}, H_\phi(\mathbf{x}_{\text{obs}}, \theta_a)) - P_\psi(\mathbf{x}_{\text{obs}}, H_\phi(\mathbf{x}_{\text{obs}}, \theta_b))\|_2 \\ &\leq \text{Lip}_P \|H_\phi(\mathbf{x}_{\text{obs}}, \theta_a) - H_\phi(\mathbf{x}_{\text{obs}}, \theta_b)\|_2.\end{aligned}$$

Applying the Lipschitz property of H_ϕ in θ yields

$$\|\mathcal{T}(\theta_a; \mathbf{x}_{\text{obs}}) - \mathcal{T}(\theta_b; \mathbf{x}_{\text{obs}})\|_2 \leq \text{Lip}_P \text{Lip}_H \|\theta_a - \theta_b\|_2.$$

Since $\text{Lip}_{\mathcal{T}} = \text{Lip}_P \text{Lip}_H < 1$, then $\mathcal{T}(\cdot; \mathbf{x}_{\text{obs}})$ is a contraction and the remaining claims follow from Banach's fixed-point theorem. ■

4.3. Enforcing Lipschitz bounds via spectral normalization. Assumption 4.1 can be promoted in practice by controlling the operator norms of the linear layers. For a linear map $x \mapsto Wx$, its ℓ_2-Lipschitz constant equals the spectral norm $\|W\|_2$. Spectral normalization rescales each weight matrix W by an estimate of $\|W\|_2$ so that the resulting layer has a prescribed operator-norm bound, typically close to 1. This idea was popularized in the deep learning literature as a training-stabilization technique. [8]

For a feed-forward network composed of linear layers and pointwise nonlinearities with Lipschitz constant at most 1 (e.g., tanh), one obtains a simple upper bound on the network's Lipschitz constant by multiplying the spectral norms of the weight matrices. Consequently, by applying spectral normalization layerwise (and leaving a small margin for strict inequality), we can ensure Lip_H and Lip_P are bounded and achieve $\mathrm{Lip}_P\mathrm{Lip}_H < 1$. Further implementation details are provided in the supplementary material.

Section 7.4 verifies this empirically: without spectral normalization the measured composite Lipschitz constant exceeds one and the iteration fails to contract, whereas across a range of margins that keep $\mathrm{Lip}_{\mathcal{T}} < 1$ the iteration converges as predicted.

4.4. Implications for the number of inference steps. Theorem 4.2 yields a direct justification for using a small, fixed number of inference steps. If $\mathcal{T}(\cdot;\mathbf{x}_{\mathrm{obs}})$ is a contraction with constant $K < 1$, then after k iterations,

$$\|\theta^{(k)} - \theta^*(\mathbf{x}_{\mathrm{obs}})\|_2 \leq K^k \|\theta^{(0)} - \theta^*(\mathbf{x}_{\mathrm{obs}})\|_2,$$

so the error decreases exponentially in k. This explains why in practice a small number of iterations (e.g., $K = 10$) suffices, and why often 1–2 iterations already yield a stable estimate.

5. Theory 2: Correctness and Statistical Limitations. Theorem 4.2 establishes that the iteration $\theta^{(k+1)} = \mathcal{T}(\theta^{(k)};\mathbf{x}_{\mathrm{obs}})$ converges to a unique fixed point $\theta^*(\mathbf{x}_{\mathrm{obs}})$ from any initialization, but convergence says nothing about *what* that fixed point is. This section characterizes it, and the conclusion is a characterization rather than a correctness guarantee. We establish four facts. First, contractivity and observational ambiguity are structurally *incompatible*: a strict contraction cannot keep two distinct admissible parameters as zero-residual fixed points, so preserving the whole teacher-forced target is unreachable on a non-identifiable fiber (Section 5.2). Second, because the fixed point is a deterministic function of $\mathbf{x}_{\mathrm{obs}}$, it cannot be more accurate than the conditional mean — the Bayes performance ceiling under squared loss faced already by a plain direct regressor, which reconstructing the hidden state does *not* circumvent (Section 5.3). Third, among all contractions the one that minimizes the idealized one-step risk has that conditional mean as its fixed point (Section 5.4); we are careful to separate this idealized objective from the decoupled loss actually optimized in training. Fourth, whether that Bayes-optimal point is itself observation-consistent depends on the *coordinate* in which the squared loss is taken: for a curved fiber the conditional mean lies off it, but a reparameterization that straightens the fiber — logarithms for a product degeneracy — keeps the conditional mean on the fiber (Section 5.5). The overall picture is that the decoupled fixed-point estimator is a stable, convergent procedure whose accuracy is bounded by the conditional mean in the loss coordinate, and whose reconstruction of the hidden state confers no *information-theoretic* advantage over an estimator based on the observation alone (in our experiments the finite-capacity direct baseline also performs better) — statements we make precise below and examine empirically in Section 7.

Loss coordinate.. The squared loss need not be taken in the physical parameter θ: training minimizes MSE in a fixed, invertible *network coordinate* $u = \tau(\theta)$ (in our OGTT pipeline τ is a componentwise logarithm; see Appendix B.2). All Bayes and fiber statements below hold in whatever coordinate carries the loss — we write them in a generic coordinate and are explicit, where it matters, about which τ is in force. Throughout, we write $m(\mathbf{x}_{\text{obs}}) := \mathbb{E}[\theta \mid \mathbf{x}_{\text{obs}}]$ for the conditional mean of the coordinate under discussion and $V(\mathbf{x}_{\text{obs}}) := \mathbb{E}\big[\|\theta - m(\mathbf{x}_{\text{obs}})\|_2^2 \mid \mathbf{x}_{\text{obs}}\big]$ for the conditional variance, and use the elementary *Bayes decomposition* of the squared risk. For any deterministic (measurable) estimator $g(\mathbf{x}_{\text{obs}})$ with $\mathbb{E}\|\theta\|_2^2 < \infty$,

$$\mathbb{E}\big\|g(\mathbf{x}_{\text{obs}}) - \theta\big\|_2^2 = \underbrace{\mathbb{E}\,V(\mathbf{x}_{\text{obs}})}_{\text{irreducible}} + \underbrace{\mathbb{E}\big\|g(\mathbf{x}_{\text{obs}}) - m(\mathbf{x}_{\text{obs}})\big\|_2^2}_{\text{estimator excess}}, \tag{5.1}$$

which follows by conditioning on $\mathbf{x}_{\text{obs}}$ and expanding the square, the cross term vanishing because $m(\mathbf{x}_{\text{obs}})$ is the conditional mean. Thus the conditional mean is the squared-risk minimizer, and $\mathbb{E}\,V$ is a floor obeyed by *every* deterministic estimator; but (5.1) does *not* say that an arbitrary deterministic fixed point equals the conditional mean — only that its excess risk over the floor is the squared distance to it.

5.1. The One-Step Residual and the Fixed-Point Error. To quantify the estimation error at the individual-sample level, we define a metric measuring how well the learned operator preserves the true parameter.

Definition 5.1 (One-Step Residual). *For a given observation $\mathbf{x}_{\text{obs}}$ and an evaluation parameter θ, the one-step residual of the iterative operator $\mathcal{T}(\cdot;\mathbf{x}_{\text{obs}})$ is*

$$\varepsilon(\mathbf{x}_{\text{obs}}, \theta) := \|\mathcal{T}(\theta;\mathbf{x}_{\text{obs}}) - \theta\|_2. \tag{5.2}$$

For an observed sample pair $(\mathbf{x}_{\text{obs}}, \theta^{\text{true}})$ drawn from the joint law — note that under non-identifiability θ^{true} is not *a function of $\mathbf{x}_{\text{obs}}$, since one observation admits many truths — we write $\varepsilon(\mathbf{x}_{\text{obs}}) := \varepsilon(\mathbf{x}_{\text{obs}}, \theta^{\text{true}})$.*

The one-step residual at the ground truth directly bounds the distance between the converged fixed point and the ground-truth parameter.

Lemma 5.2 (Fixed-Point Error Bound). *Suppose $\mathcal{T}(\cdot;\mathbf{x}_{\text{obs}})$ is a contraction mapping with a Lipschitz constant $\mathrm{Lip}_{\mathcal{T}} < 1$, and let $\theta^* = \theta^*(\mathbf{x}_{\text{obs}})$ be its unique fixed point. Then the distance between the fixed point and the ground-truth parameter is bounded by the one-step residual:*

$$\|\theta^* - \theta^{\text{true}}\|_2 \le \frac{\varepsilon(\mathbf{x}_{\text{obs}})}{1 - \mathrm{Lip}_{\mathcal{T}}}. \tag{5.3}$$

Proof. By the triangle inequality and the contractivity of $\mathcal{T}(\cdot;\mathbf{x}_{\text{obs}})$,

$$\begin{aligned}
\|\theta^* - \theta^{\text{true}}\|_2 &= \|\mathcal{T}(\theta^\star;\mathbf{x}_{\text{obs}}) - \theta^{\text{true}}\|_2 \\
&\le \|\mathcal{T}(\theta^*;\mathbf{x}_{\text{obs}}) - \mathcal{T}(\theta^{\text{true}};\mathbf{x}_{\text{obs}})\|_2 + \|\mathcal{T}(\theta^{\text{true}};\mathbf{x}_{\text{obs}}) - \theta^{\text{true}}\|_2 \\
&\le \mathrm{Lip}_{\mathcal{T}}\|\theta^* - \theta^{\text{true}}\|_2 + \varepsilon(\mathbf{x}_{\text{obs}}).
\end{aligned}$$

Rearranging yields the bound. ∎

Teacher forcing *targets* $\varepsilon(\mathbf{x}_{\text{obs}}) = 0$: during training, H_ϕ and P_ψ are optimized on the true pairs so that $H_\phi(\mathbf{x}_{\text{obs}}, \theta^{\text{true}}) \approx \mathbf{x}_{\text{hid}}$ and $P_\psi(\mathbf{x}_{\text{obs}}, \mathbf{x}_{\text{hid}}) \approx \theta^{\text{true}}$, which would make $\mathcal{T}(\theta^{\text{true}}; \mathbf{x}_{\text{obs}}) \approx \theta^{\text{true}}$ and, by Lemma 5.2, pin the fixed point to the ground truth. The next subsection shows that this ideal is, under observational ambiguity, not merely hard to reach but structurally *unreachable*.

5.2. Contraction and Ambiguity Are Incompatible. The following is our central negative result. It is a purely geometric consequence of contractivity, independent of how the networks are trained: it lower-bounds the one-step residual by the distance to the operator's own fixed point, and hence forbids a strict contraction from preserving two distinct admissible parameters.

Theorem 5.3 (Contraction–ambiguity incompatibility). *Fix $\mathbf{x}_{\text{obs}}$ and suppose $\mathcal{T}(\cdot; \mathbf{x}_{\text{obs}})$ is a q-contraction ($q := \mathrm{Lip}_{\mathcal{T}} < 1$) with unique fixed point $g(\mathbf{x}_{\text{obs}})$. Then for every $\theta \in \Theta$,*

$$\varepsilon(\mathbf{x}_{\text{obs}}, \theta) \;\geq\; (1-q)\, \big\|\theta - g(\mathbf{x}_{\text{obs}})\big\|_2. \tag{5.4}$$

Consequently, taking the conditional expectation over $\theta \mid \mathbf{x}_{\text{obs}}$ and using (5.1),

$$\mathbb{E}\big[\varepsilon(\mathbf{x}_{\text{obs}}, \theta)^2 \mid \mathbf{x}_{\text{obs}}\big] \;\geq\; (1-q)^2 \Big(V(\mathbf{x}_{\text{obs}}) + \big\|g(\mathbf{x}_{\text{obs}}) - m(\mathbf{x}_{\text{obs}})\big\|_2^2\Big) \;\geq\; (1-q)^2\, V(\mathbf{x}_{\text{obs}}). \tag{5.5}$$

Moreover, for any two parameters $\theta_a \neq \theta_b$ consistent with the same observation,

$$\varepsilon(\mathbf{x}_{\text{obs}}, \theta_a) + \varepsilon(\mathbf{x}_{\text{obs}}, \theta_b) \;\geq\; (1-q)\, \|\theta_a - \theta_b\|_2. \tag{5.6}$$

Proof. Write $g = g(\mathbf{x}_{\text{obs}})$ and $\mathcal{T} = \mathcal{T}(\cdot; \mathbf{x}_{\text{obs}})$. Since $\mathcal{T}(g) = g$, the reverse triangle inequality gives

$$\varepsilon(\mathbf{x}_{\text{obs}}, \theta) = \|\mathcal{T}(\theta) - \theta\|_2 \geq \|\theta - g\|_2 - \|\mathcal{T}(\theta) - g\|_2 = \|\theta - g\|_2 - \|\mathcal{T}(\theta) - \mathcal{T}(g)\|_2 \geq (1-q)\|\theta - g\|_2,$$

which is (5.4). Squaring and taking $\mathbb{E}[\cdot \mid \mathbf{x}_{\text{obs}}]$, then applying the bias–variance identity $\mathbb{E}[\|\theta - g\|_2^2 \mid \mathbf{x}_{\text{obs}}] = V(\mathbf{x}_{\text{obs}}) + \|g - m(\mathbf{x}_{\text{obs}})\|_2^2$ (a pointwise instance of (5.1)), gives (5.5). For (5.6), the triangle inequality and contractivity give

$$\|\theta_a - \theta_b\|_2 \leq \|\theta_a - \mathcal{T}(\theta_a)\|_2 + \|\mathcal{T}(\theta_a) - \mathcal{T}(\theta_b)\|_2 + \|\mathcal{T}(\theta_b) - \theta_b\|_2 \leq \varepsilon_a + q\|\theta_a - \theta_b\|_2 + \varepsilon_b,$$

with $\varepsilon_a := \varepsilon(\mathbf{x}_{\text{obs}}, \theta_a)$, $\varepsilon_b := \varepsilon(\mathbf{x}_{\text{obs}}, \theta_b)$; rearranging yields (5.6). ∎

Theorem 5.3 makes the tension explicit. A single observation with two admissible ground truths $\theta_a \neq \theta_b$ cannot have both as zero-residual points of a strict contraction: by (5.6) their residuals sum to at least $(1-q)\|\theta_a - \theta_b\| > 0$. The teacher-forced ideal — preserve *every* admissible ground truth, $\varepsilon = 0$ over the whole support — is therefore incompatible with global strict contractivity on a non-identifiable fiber. The incompatibility is a *population* statement: a finite random training set drawn from a continuous prior rarely contains the same $\mathbf{x}_{\text{obs}}$ twice, so a network can interpolate nearby fiber observations at training time; the contradiction surfaces at the population/generalization level, where the operator must act on the whole fiber at once. The residual it must then leave behind is not a training imperfection but a structural floor: by (5.5) the expected squared residual is bounded below by $(1-q)^2\,V(\mathbf{x}_{\text{obs}})$, strictly positive exactly when the observation fails to identify θ.

Two *complementary* floors follow, and it is worth keeping their logic straight. The residual floor above is a lower bound on $\mathbb{E}[\varepsilon^2 \mid \mathbf{x}_{\text{obs}}]$; Lemma 5.2, by contrast, runs the other way — it converts a *measured* residual into an *upper* certificate for a particular fixed-point error, and does not by itself lower-bound that error. The unavoidable lower bound on every deterministic fixed point comes independently from the Bayes decomposition (Proposition 5.4 below). Finally, the residual floor $(1-q)^2 V$ shrinks as $q \uparrow 1$ and vanishes in the limit — but only at the price of arbitrarily slow convergence, and any *fixed* strict margin $q < 1$ keeps it positive; the fixed-point Bayes floor $\mathbb{E}\,V$ itself is independent of q.

5.3. The Conditional-Mean Ceiling: Reconstructing the Hidden State Adds No Information. A natural intuition motivates the decoupled design: since different hidden states and parameters can produce the same observation, a method that reconstructs the hidden state "ought" to have an information advantage over a direct regressor $\mathbf{x}_{\text{obs}} \mapsto \theta$ that ignores it. The following proposition shows this intuition is mistaken, because at inference the reconstructed hidden state is itself a function of $\mathbf{x}_{\text{obs}}$.

Proposition 5.4 (Conditional-mean ceiling). *By uniqueness of the fixed point (Theorem 4.2), $\theta^*(\mathbf{x}_{\text{obs}})$ is a deterministic function of $\mathbf{x}_{\text{obs}}$; it is moreover measurable, since $\mathcal{T}$ is continuous in $(\mathbf{x}_{\text{obs}}, \theta)$ with a contraction constant uniform in $\mathbf{x}_{\text{obs}}$, so $\theta^*(\mathbf{x}_{\text{obs}}) = \lim_k \mathcal{T}^{(k)}(\theta^{(0)}; \mathbf{x}_{\text{obs}})$ is a pointwise limit of continuous (hence measurable) maps. Consequently:*

(i) the reconstructed hidden state at convergence, $H_\phi\big(\mathbf{x}_{\text{obs}}, \theta^*(\mathbf{x}_{\text{obs}})\big)$, is also a function of $\mathbf{x}_{\text{obs}}$, and hence carries no information about θ beyond what $\mathbf{x}_{\text{obs}}$ already contains;

(ii) by the Bayes decomposition (5.1), the expected squared error is bounded below by the expected conditional variance,

$$\mathbb{E}\big\|\theta^*(\mathbf{x}_{\text{obs}}) - \theta\big\|_2^2 \;\geq\; \mathbb{E}\big[V(\mathbf{x}_{\text{obs}})\big] = \mathbb{E}\big[\operatorname{Var}(\theta \mid \mathbf{x}_{\text{obs}})\big],$$

the same Bayes floor obeyed by any *estimator depending on $\mathbf{x}_{\text{obs}}$ alone — in particular by direct regression — and attained only by the conditional mean $m(\mathbf{x}_{\text{obs}})$.*

Proof. Uniqueness makes $\theta^* = g(\mathbf{x}_{\text{obs}})$ for a measurable g, giving (i) by composition. Part (ii) is (5.1) applied to $g = \theta^*$, discarding the nonnegative excess term, with equality iff $g = m$ almost surely. ∎

Proposition 5.4 identifies the fundamental ceiling. Teacher forcing exploits the *true* hidden state during training, but that advantage does not transfer to inference, where the true hidden state is unavailable and its learned surrogate is a deterministic image of $\mathbf{x}_{\text{obs}}$. The decoupled iteration therefore cannot beat the conditional mean, and — as our experiments show (Section 7) — a direct regressor typically *reaches* that ceiling more closely, since it avoids the reconstruction detour that inflates ε. The value of the fixed-point construction thus lies in its convergence guarantee and in the self-consistent hidden trajectory it returns as a by-product, not in improved parameter accuracy.

5.4. Which Contraction? A Variational Route to the Conditional-Mean Fixed Point. Theorem 5.3 lower-bounds the residual of *any* contraction; we now ask which contraction is best in the idealized sense of minimizing the expected one-step residual, and show that its fixed point is forced to be the conditional mean. This derives the conditional-mean collapse from an optimization principle rather than assuming it. We first recall the classical squared-loss fact for context.

Proposition 5.5 (Optimal predictor under squared loss). *Let* $(\mathbf{x}_{\text{obs}}, \theta)$ *be drawn from* $p(\mathbf{x}_{\text{obs}}, \theta)$ *with* $\mathbb{E}[\|\theta\|_2^2] < \infty$. *Among all measurable* $g(\mathbf{x}_{\text{obs}})$ *minimizing* $\mathbb{E}\,\|\theta - g(\mathbf{x}_{\text{obs}})\|_2^2$, *the minimizer is the conditional mean* $g^*(\mathbf{x}_{\text{obs}}) = m(\mathbf{x}_{\text{obs}})$ *almost surely.*

This concerns an *unconstrained* regressor. Our operator is not unconstrained: it must be a contraction in θ. We therefore pose the idealized operator-learning problem of choosing, for each $\mathbf{x}_{\text{obs}}$ and a fixed contraction budget $q < 1$, the q-contraction $T_{\mathbf{x}_{\text{obs}}}(\cdot)$ on the convex set Θ that minimizes the conditional one-step risk

$$\mathcal{R}_{\mathbf{x}_{\text{obs}}}(T) := \mathbb{E}\big[\, \|T_{\mathbf{x}_{\text{obs}}}(\theta) - \theta\|_2^2 \,\big|\, \mathbf{x}_{\text{obs}}\big]. \tag{5.7}$$

Theorem 5.6 (Risk-optimal contraction and its fixed point). *Assume* $\Theta \subseteq \mathbb{R}^p$ *is convex and closed,* $m(\mathbf{x}_{\text{obs}}) \in \Theta$, *and fix* $q \in [0, 1)$. *Then:*

(i) every q*-contraction* T *obeys* $\mathcal{R}_{\mathbf{x}_{\text{obs}}}(T) \geq (1-q)^2\, V(\mathbf{x}_{\text{obs}})$, *by* (5.5)*;*

(ii) the affine contraction $T^{\text{opt}}_{\mathbf{x}_{\text{obs}}}(\theta) = q\,\theta + (1-q)\, m(\mathbf{x}_{\text{obs}})$ *is a* q*-contraction mapping* Θ *into* Θ, *and attains the bound:* $\mathcal{R}_{\mathbf{x}_{\text{obs}}}(T^{\text{opt}}) = (1-q)^2\, V(\mathbf{x}_{\text{obs}})$*;*

(iii) the minimum risk is $(1-q)^2\, V(\mathbf{x}_{\text{obs}})$, *and any minimizer has the conditional mean* $m(\mathbf{x}_{\text{obs}})$ *as its fixed point.*

Proof. (i) is (5.5). For (ii), T^{opt} has Lipschitz constant $q < 1$, and for $\theta \in \Theta$ it is the convex combination $q\theta + (1-q)m(\mathbf{x}_{\text{obs}})$ of two points of Θ, hence lies in Θ; its unique fixed point is $m(\mathbf{x}_{\text{obs}})$. Its residual is $T^{\text{opt}}(\theta) - \theta = (1-q)\big(m(\mathbf{x}_{\text{obs}}) - \theta\big)$, so $\mathcal{R}_{\mathbf{x}_{\text{obs}}}(T^{\text{opt}}) = (1-q)^2\,\mathbb{E}[\|\theta - m(\mathbf{x}_{\text{obs}})\|_2^2 \mid \mathbf{x}_{\text{obs}}] = (1-q)^2 V(\mathbf{x}_{\text{obs}})$. Combining (i) and (ii) gives the minimum value in (iii). Finally, if T is any q-contraction with fixed point g, then (5.5) gives $\mathcal{R}_{\mathbf{x}_{\text{obs}}}(T) \geq (1-q)^2(V(\mathbf{x}_{\text{obs}}) + \|g - m(\mathbf{x}_{\text{obs}})\|_2^2)$; attaining the minimum $(1-q)^2 V(\mathbf{x}_{\text{obs}})$ forces $\|g - m(\mathbf{x}_{\text{obs}})\|_2 = 0$, i.e. $g = m(\mathbf{x}_{\text{obs}})$. ■

Remark 5.7 (What is and is not optimized in training). Theorem 5.6 concerns the *composite* one-step risk (5.7) of an idealized contractive operator. The decoupled teacher forcing of Section 3.3 does *not* minimize (5.7): it minimizes the two separate losses $\mathcal{L}_H + \mathcal{L}_P$ on ground-truth pairs, with no term tying $P_\psi \circ H_\phi$ to the identity in the composed sense. We therefore do *not* claim that the trained operator equals the risk-optimal contraction T^{opt}, nor that it exactly realizes the conditional-mean fixed point. Theorem 5.6 should be read as a statement about (a) the idealized contraction that best trades off residual against ambiguity, and (b) the explicit consistency objective

$$\mathcal{L}_T := \mathbb{E}\,\big\|P_\psi\big(\mathbf{x}_{\text{obs}}, H_\phi(\mathbf{x}_{\text{obs}}, \theta)\big) - \theta\big\|_2^2, \tag{5.8}$$

which is a sample form of (5.7). The conditional-mean collapse observed for the trained decoupled operator (Section 7) is then reported as *empirical* corroboration of this theoretical prediction, not as an identity guaranteed by the training loss.

We record the population consequences used in the sequel, separating the risk-optimal case (an equality) from an arbitrary learned contraction (a sandwich).

Theorem 5.8 (Conditional-mean operator: residual and error are exact). *Suppose the operator's fixed point is the conditional mean, $g = m(\mathbf{x}_{\text{obs}})$ — attained, for any $q \in [0,1)$, by the risk-optimal affine contraction T^{opt} of Theorem 5.6, and in particular by the constant map $\mathcal{T}(\theta;\mathbf{x}_{\text{obs}}) = m(\mathbf{x}_{\text{obs}})$. Then the fixed-point error is* exactly *the Bayes floor,*

$$\mathbb{E}\big\|\theta^*(\mathbf{x}_{\text{obs}}) - \theta\big\|_2^2 = \mathbb{E}_{\mathbf{x}_{\text{obs}}}\big[\operatorname{Var}(\theta \mid \mathbf{x}_{\text{obs}})\big], \tag{5.9}$$

and the expected squared one-step residual at the ground truth is $\mathbb{E}[\varepsilon(\mathbf{x}_{\text{obs}})^2] = (1-q)^2\,\mathbb{E}[\operatorname{Var}(\theta \mid \mathbf{x}_{\text{obs}})]$ for T^{opt} (equal to $\mathbb{E}[\operatorname{Var}(\theta \mid \mathbf{x}_{\text{obs}})]$ for the constant map).

Proof. Since $\theta^* = g = m(\mathbf{x}_{\text{obs}})$, the excess term in the Bayes decomposition (5.1) vanishes, giving the first equality. For T^{opt}, $\varepsilon(\mathbf{x}_{\text{obs}}) = \|T^{\text{opt}}(\theta^{\text{true}}) - \theta^{\text{true}}\|_2 = (1-q)\|m(\mathbf{x}_{\text{obs}}) - \theta^{\text{true}}\|_2$; squaring and taking $\mathbb{E}[\,\cdot \mid \mathbf{x}_{\text{obs}}]$ then the tower rule gives the residual identity. ∎

Corollary 5.9 (Regression-to-the-mean sandwich). *For* any *learned q-contraction ($q = \operatorname{Lip}_{\mathcal{T}} < 1$) with fixed point θ^*, the Bayes lower bound (Proposition 5.4) and the residual upper bound (Lemma 5.2) sandwich the fixed-point risk:*

$$\mathbb{E}_{\mathbf{x}_{\text{obs}}}\big[\operatorname{Var}(\theta \mid \mathbf{x}_{\text{obs}})\big] \;\le\; \mathbb{E}\big\|\theta^*(\mathbf{x}_{\text{obs}}) - \theta\big\|_2^2 \;\le\; \frac{\mathbb{E}\big[\varepsilon(\mathbf{x}_{\text{obs}})^2\big]}{(1-q)^2}. \tag{5.10}$$

The lower bound is attained iff $\theta^* = m$ (Theorem 5.8); it is strictly positive whenever $\operatorname{Var}(\theta \mid \mathbf{x}_{\text{obs}}) > 0$.

Corollary 5.9 shows that even under perfect, stable convergence ($\operatorname{Lip}_{\mathcal{T}} < 1$), the fixed point cannot align with the sample-specific truth whenever $\operatorname{Var}(\theta \mid \mathbf{x}_{\text{obs}}) > 0$: its risk is floored at the Bayes level, and the risk-optimal contraction attains that floor exactly by collapsing to the conditional mean. This is the regression-to-the-mean and joint-distribution collapse of deterministic squared-loss estimators under non-identifiability.

5.5. The Conditional Mean Lies Off the Solution Fiber. Corollary 5.9, together with the variational route of Theorem 5.6, identifies the conditional mean $m(\mathbf{x}_{\text{obs}})$ as the relevant target. We now sharpen this into a geometric statement: under a curved non-identifiability, the conditional mean is not merely inaccurate — it is not itself a parameter consistent with the observation, because it lies strictly *off* the solution fiber. We are explicit about the conditions this requires.

Assumption 5.10 (Fiber model). For a given observation, write $c = c(\mathbf{x}_{\text{obs}})$ for the value of the identifiable statistic and let the *solution fiber* be the set of parameters consistent with the observation,

$$\mathcal{F}_c = \{\theta \in \Theta : \theta \text{ reproduces } \mathbf{x}_{\text{obs}}\}.$$

We assume (i) the data are noiseless and generated by the model, and any nuisance initial conditions are marginalized under the prior π, so that the support of the conditional law $p(\theta \mid \mathbf{x}_{\text{obs}})$ is $\mathcal{F}_c \cap \operatorname{supp}(\pi)$ (a subset of the physical fiber, and equal to it when π has full support); and (ii) the conditional law is not a Dirac measure — equivalently, its support contains at least two points; for a continuous prior this means it assigns positive probability to two disjoint relative neighborhoods on the fiber.

Under Assumption 5.10, the following statement is geometric and does not depend on the specific form of the degeneracy: whenever the fiber is genuinely curved *in the coordinate carrying the squared loss*, the conditional mean of that coordinate leaves it. Curvature is coordinate-dependent, and this dependence is the crux of the OGTT discussion below.

Proposition 5.11 (Conditional mean off a curved fiber). *Suppose Assumption 5.10 holds and the fiber $\mathcal{F}_c \subset \mathbb{R}^p$ is compact and in* strictly convex position*: every point of $\mathcal{F}_c$ is an extreme point of its convex hull* $\mathrm{conv}(\mathcal{F}_c)$*. Then*

$$m(\mathbf{x}_{\mathrm{obs}}) = \mathbb{E}[\theta \mid \mathbf{x}_{\mathrm{obs}}] \notin \mathcal{F}_c.$$

Hence an estimator attaining the population squared-loss Bayes rule (Proposition 5.5) returns the conditional mean, which is observation-inconsistent: it lies off the solution fiber.

Proof. Let $m = m(\mathbf{x}_{\mathrm{obs}})$ be the barycenter of the conditional law. Since that law is supported on $\mathcal{F}_c \subseteq \mathrm{conv}(\mathcal{F}_c)$, we have $m \in \mathrm{conv}(\mathcal{F}_c)$. By strictly-convex position, $\mathcal{F}_c$ is the set of extreme points of the compact convex body $\mathrm{conv}(\mathcal{F}_c)$. By Bauer's characterization of extreme points via representing measures [10], the barycenter of a probability measure supported on such a body is an extreme point e if and only if the measure is the Dirac mass δ_e. As the conditional law is non-degenerate (Assumption 5.10(ii)), m is not an extreme point; hence $m \notin \mathcal{F}_c$. ■

The 1-D product degeneracy is the canonical instance of a curved physical fiber, and there the displacement under a *physical-coordinate* squared loss is fully explicit.

Corollary 5.12 (Product degeneracy under physical-coordinate MSE). *Suppose the observation identifies only the product $c = \theta_1\theta_2$ on the physical domain $\theta_1, \theta_2 > 0$ (the constant-HGP glucose-only OGTT idealization is exactly of this form, and the sparse full-HGP regime approximately so; see Section 6), so that on a bounded parameter box the fiber is the hyperbolic arc $\mathcal{F}_c = \{(\theta_1, c/\theta_1)\}$. Under squared loss* in the physical coordinate (θ_1, θ_2)*, $\mathcal{F}_c$ is in strictly convex position, so Proposition 5.11 applies, and the bias is quantified by*

$$\mathbb{E}[\theta_1 \mid c]\, \mathbb{E}[\theta_2 \mid c] = c\, \mathbb{E}[\theta_1 \mid c]\, \mathbb{E}[\theta_1^{-1} \mid c] \;\geq\; c,$$

with strict inequality unless θ_1 is almost surely constant given c. The product of conditional means thus exceeds the true identifiable value c: the conditional mean lies strictly on the convex side of the fiber.

Proof. The map $\theta_1 \mapsto c/\theta_1$ is strictly convex on $(0, \infty)$, so the hyperbolic arc lies strictly on one side of each of its chords; every point is therefore extreme in $\mathrm{conv}(\mathcal{F}_c)$, giving strictly convex position. For the bound, $\theta_2 = c/\theta_1$ on $\mathcal{F}_c$ gives $\mathbb{E}[\theta_2 \mid c] = c\,\mathbb{E}[\theta_1^{-1} \mid c]$, and Jensen's inequality for the strictly convex $x \mapsto x^{-1}$ yields $\mathbb{E}[\theta_1^{-1} \mid c] \geq (\mathbb{E}[\theta_1 \mid c])^{-1}$, with equality iff θ_1 is a.s. constant. Multiplying by $\mathbb{E}[\theta_1 \mid c] > 0$ gives the claim. ■

Remark 5.13 (The off-fiber displacement is a property of the loss coordinate). Corollary 5.12 is stated for squared loss in the physical coordinate (θ_1, θ_2). The displacement is *not* intrinsic to the product degeneracy: it is a property of that coordinate. Under the logarithmic reparameterization $u = (\log\theta_1, \log\theta_2)$ the same fiber becomes the *affine* line $u_1 + u_2 = \log c$, which is not curved, so its (log-coordinate) conditional mean stays on it,

$$\mathbb{E}[\log\theta_1 \mid c] + \mathbb{E}[\log\theta_2 \mid c] = \log c \quad\Longrightarrow\quad e^{\mathbb{E}[\log\theta_1|c]}\, e^{\mathbb{E}[\log\theta_2|c]} = c,$$

i.e. the back-transformed conditional geometric mean reproduces the product exactly. Whether the Bayes-optimal point leaves the fiber thus depends on the loss coordinate: physical-coordinate MSE incurs the arithmetic-mean Jensen bias above, whereas log-coordinate MSE — the coordinate our OGTT pipeline actually uses (Appendix B.2) — does not. Corollary 5.12 should therefore be read as an exact theoretical example of physical-coordinate MSE, *not* as a description of the log-coordinate OGTT experiments; we make this distinction operational in Section 7.3.

The scope of Proposition 5.11 is therefore precise: it does not assert that *every* squared-loss estimator lies off the fiber, but that an estimator attaining the population squared-loss Bayes rule in a coordinate where the fiber is curved returns a conditional mean off that fiber (Assumption 5.10). For our fixed-point operator this bites only to the extent that (a) its fixed point is close to the relevant conditional mean — a link supported theoretically by the variational Theorem 5.6 and, for the trained operator, empirically in Section 7 — and (b) the loss coordinate makes the fiber curved. Because the OGTT pipeline trains in log coordinates, its product fiber is affine and the off-fiber arithmetic-mean bias of Corollary 5.12 does not apply to it; the residual product error we observe there is attributed to operator approximation, not to an off-fiber conditional mean (Section 7.3).

Remark 5.14 (Two sources of the fixed-point error). The one-step residual $\varepsilon(\mathbf{x}_{\text{obs}})$ governing Lemma 5.2 conflates two contributions: (i) a *structural* part, bounded below by $(1-q)^2V(\mathbf{x}_{\text{obs}})$ (Theorem 5.3); for the risk-optimal affine contraction its expected square is exactly $(1-q)^2V(\mathbf{x}_{\text{obs}})$ (equal to V only in the $q \to 0$ constant-map limit; Theorem 5.8), irreducible under non-identifiability; and (ii) an *operator-approximation* part that persists even when the observation is fully identifying ($V(\mathbf{x}_{\text{obs}}) = 0$), reflecting the finite capacity and decoupled teacher-forced training of the learned composition $P_\psi \circ H_\phi$. The two can be contrasted diagnostically: an identifiable, well-conditioned system (SIR) exposes (ii) with the structural term small, whereas a non-identifiable system is dominated by (i). Section 7 instantiates both regimes.

6. Feasibility: Identifiability and Observability Analysis.

Algorithmic convergence of the fixed-point iteration guarantees mathematical stability, but it does not ensure correct parameter recovery if the underlying inverse problem is structurally ill-posed. In this section, we analyze the structural identifiability of our benchmark systems using the nonlinear observability framework of Hermann and Krener [5].

6.1. Nonlinear Observability Framework.

We formulate parameter identifiability as a state observability problem by augmenting the state space. Let $\mathbf{x} \in \mathbb{R}^d$ be the system state and $\theta \in \mathbb{R}^p$ be the constant unknown parameters. We define the augmented state $\mathbf{z} = [\mathbf{x}^\top, \theta^\top]^\top \in \mathcal{M} \subseteq \mathbb{R}^m$, where $m = d + p$. The augmented dynamics and observation map are:

$$\dot{\mathbf{z}} = F(\mathbf{z}) = \begin{bmatrix} f(\mathbf{x},\theta) \\ \mathbf{0} \end{bmatrix}, \quad \mathbf{x}_{obs} = \tilde{h}(\mathbf{z}) = h(\mathbf{x}). \tag{6.1}$$

Following Hermann and Krener, we construct the observability map $\Phi(\mathbf{z}) : \mathcal{M} \to \mathbb{R}^{m\times d_{obs}}$ using successive Lie derivatives of the observation function $\tilde{h}$ along the vector field F:

$$\Phi(\mathbf{z}) = \begin{bmatrix} \tilde{h}(\mathbf{z}) \\ L_F\tilde{h}(\mathbf{z}) \\ \vdots \\ L_F^{m-1}\tilde{h}(\mathbf{z}) \end{bmatrix}, \tag{6.2}$$

where $L_F\tilde{h}(\mathbf{z}) = \nabla_{\mathbf{z}}\tilde{h}(\mathbf{z}) \cdot F(\mathbf{z})$. The local observability matrix is defined as the Jacobian $\mathbf{O}(\mathbf{z}) := \nabla_{\mathbf{z}}\Phi(\mathbf{z})$. If $\mathbf{O}(\mathbf{z})$ has full column rank m at a generic $\mathbf{z}$, the system satisfies the observability rank condition: the map Φ is then a local immersion (an inverse-function argument on any square submap with nonvanishing minor recovers $\mathbf{z}$ locally), so the augmented state $\mathbf{z}$ — including θ — is *locally weakly observable*, i.e. locally identifiable at generic states. This is a local, generic statement; it does not by itself assert global uniqueness.

Two cautions frame how we use this criterion. First, the Hermann–Krener rank condition is a statement about the *continuous-time, exact* output $h(\mathbf{x}(t))$ and its derivatives $L_F^k\tilde{h}$: it certifies *local weak observability* in an idealized information limit and is silent about how much of that information survives finite, noisy, sparse sampling. To make the practical question precise we introduce the *sampled observation map*

$$\mathcal{G}(\mathbf{z}_0) = \big(h(\mathbf{x}(t_0)),\, h(\mathbf{x}(t_1)),\, \ldots,\, h(\mathbf{x}(t_N))\big), \tag{6.3}$$

whose Jacobian $\nabla_{\mathbf{z}_0}\mathcal{G}$ we analyze by its rank *and its conditioning* (singular-value spectrum) at the actual sampling grid. A system can be locally observable in the sense of $\mathbf{O}(\mathbf{z})$ yet have a sampled Jacobian $\nabla_{\mathbf{z}_0}\mathcal{G}$ so ill-conditioned that its inverse is unusable at any realistic noise or precision level — *practically* non-identifiable, even though it is not exactly rank-deficient. Second, full rank guarantees local reconstructability of the state, not a well-conditioned inverse; the gap between the two is exactly the ill-conditioning we quantify below. We therefore treat continuous-time observability and sparse-sampling conditioning as two distinct diagnostics, and corroborate both against the empirical direct-regression recovery of Section 7.

6.2. SIR: Exact Observability via Lie Derivatives. For the SIR compartmental model of epidemic dynamics [6], the recovered population R is strictly determined by the conservation law ($N = S + I + R$). Thus, the effective augmented state is $\mathbf{z} = [S, I, \beta, \gamma]^\top \in \mathbb{R}^4$. Under the partial observation setting, only the susceptible population is observed, yielding $\tilde{h}(\mathbf{z}) = S$.

We explicitly construct the 4×4 observability matrix by computing the sequence of Lie derivatives along the augmented vector field $F(\mathbf{z}) = [-\beta\frac{SI}{N}, \beta\frac{SI}{N} - \gamma I, 0, 0]^\top$. The first two gradient vectors are derived as:

$$\nabla_{\mathbf{z}}L_F^0\tilde{h}(\mathbf{z}) = [1, 0, 0, 0], \tag{6.4}$$

$$\nabla_{\mathbf{z}}L_F^1\tilde{h}(\mathbf{z}) = \left[-\frac{\beta I}{N}, -\frac{\beta S}{N}, -\frac{SI}{N}, 0\right]. \tag{6.5}$$

Continuing this process up to $L_F^3\tilde{h}(\mathbf{z})$ systematically embeds the nonlinear interactions of $S, I, \beta,$ and γ into the rows of $\mathbf{O}(\mathbf{z})$. Evaluating the determinant of the resulting matrix confirms that rank $\mathbf{O}(\mathbf{z}) = 4$ for generic states $\mathbf{z}$ (the closed-form determinant is $\det \mathbf{O}_S = I^3 S^4 \beta^4 / N^5$; Appendix A.1). By the rank condition this makes the SIR system *locally weakly observable* — the parameters are locally identifiable at generic states — from $S(t)$ alone. The complete step-by-step derivation of the observability matrix is provided in Appendix A.1.

6.3. Lotka-Volterra: Unidentifiability via Symbolic Arithmetic. For the Lotka–Volterra predator–prey system [7, 14], the augmented state expands to dimension $m = 6$, given by $\mathbf{z} = [x, y, \alpha, \beta, \delta, \gamma]^\top$. The observation function isolates the prey population, $\tilde{h}(\mathbf{z}) = x$.

Constructing the corresponding 6×6 observability matrix $\mathbf{O}(\mathbf{z})$ requires evaluating up to the 5th-order Lie derivative ($L_F^5 \tilde{h}$). Due to the exponential growth of polynomial cross-terms, manual derivation becomes intractable. Therefore, we utilize symbolic algebraic computation (via SymPy) to generate the gradients and assemble $\mathbf{O}(\mathbf{z})$. The symbolic evaluation analytically confirms that:

$$\operatorname{rank} \mathbf{O}(\mathbf{z}) = 5 < 6. \tag{6.6}$$

Because the algebraic rank is strictly less than the dimension of the augmented state space, the system is not locally observable from prey-only observations. The deficient direction is an explicit one-parameter scaling symmetry of the prey channel: since $\dot{x} = \alpha x - \beta x y$ depends on y only through βy, the joint rescaling

$$y(t) \mapsto c\, y(t), \qquad y_0 \mapsto c\, y_0, \qquad \beta \mapsto \beta / c \tag{6.7}$$

leaves $x(t)$ unchanged (Figure 3). The null direction is therefore the *joint* (y_0, β) scaling, not β alone; β is recovered up to the unknown hidden initial predator y_0. In our experiments y_0 is a randomized hidden nuisance, so marginalizing over it is what collapses the recovery of β specifically — if y_0 were known, β would be identifiable. The SymPy code for the rank derivation is in Appendix A.2.

6.4. OGTT: Sampled-Sensitivity Rank and Conditioning. For the OGTT (oral glucose tolerance test) glucose–insulin ISS model [3, 4], the full augmented state $\mathbf{z} = [G, I, N_5, N_6, S_I, \sigma]^\top$ is six-dimensional, but deriving its Hermann–Krener matrix symbolically is intractable and we do not attempt it. Instead we analyze the identifiability that the *actual* sparse record supports. The observed record is glucose at the five-point grid $t \in \{0, 30, 60, 90, 120\}$; the generator draws G_0, I_0, S_I, σ and slaves the incretin initial states to the steady state $(N_{5,0}, N_{6,0}) = \mathrm{SS}(G_0, \sigma)$ (Appendix B.1). Since $G_0 = G(t_0)$ is directly observed, the quantities a glucose-only record can determine are the hidden initial insulin I_0 and the parameters, i.e. the free-variable manifold

$$\xi = (I_0, S_I, \sigma) \in \mathbb{R}^3, \qquad \mathcal{G}(\xi) = \big(G(t_0), \ldots, G(t_4)\big) \in \mathbb{R}^5.$$

We study the *log-relative* sampled-sensitivity Jacobian $\mathbf{J} = \nabla_\xi \mathcal{G} \operatorname{diag}(\xi) \in \mathbb{R}^{5\times 3}$ (columns $\partial G / \partial \log \xi_k$, glucose in mg/dL), evaluated at the nominal point $G_0 = 91$ mg/dL, $\xi = (I_0, S_I, \sigma) = (6, 0.5, 0.5)$ by step-size-stable central differences (Appendix A.3). This is a scale-invariant *relative*-coordinate diagnostic, distinct from the network training coordinate of Appendix B.2; it has at most $\min(5, 3) = 3$ singular values, and since $G(t_0) = G_0$ is fixed the first row is

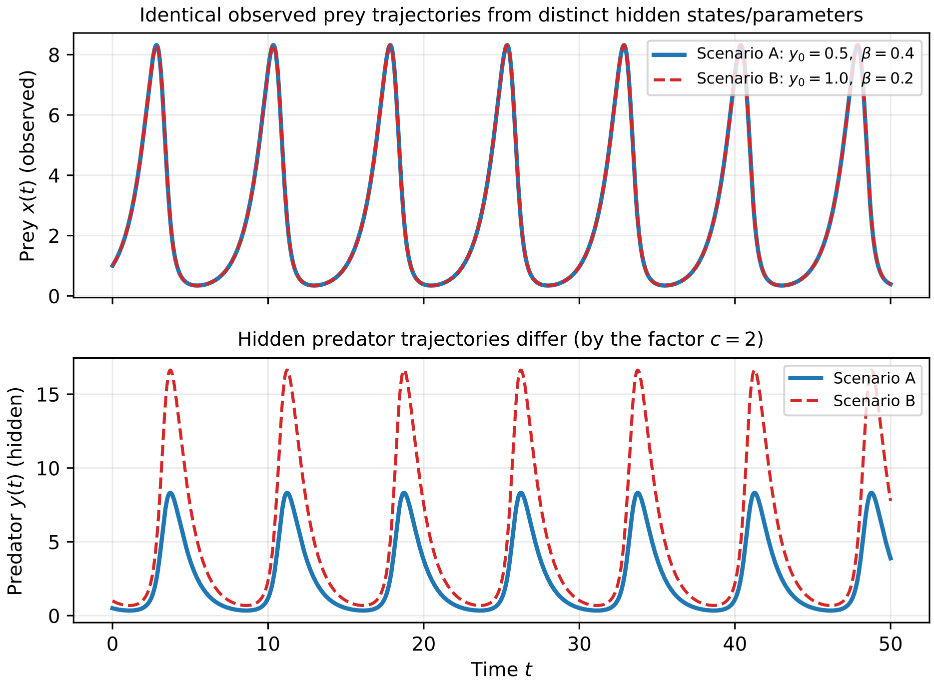


Figure 3: Structural unidentifiability in the Lotka-Volterra model. Scenario A ($x_0 = 1.0, y_0 = 0.5, \beta = 0.4$) and Scenario B ($x_0 = 1.0, y_0 = 1.0, \beta = 0.2$) produce identical observable trajectories for the prey $\mathbf{x}_{obs}(t)$ (top), despite originating from completely different hidden predator states and parameters (bottom). This algebraic indistinguishability proves that the full system cannot be identified from partial observations alone.

identically zero (four informative rows). We write ς for singular values to avoid collision with σ.

At the five-point grid the SVD gives $\varsigma \approx (73, 0.16, 0.073)$ (condition number $\approx 10^3$; Figure 4, left). Glucose is nearly equally sensitive to S_I and σ individually (log-relative column norms ≈ 51 each) with almost identical temporal profiles (Figure 4, right), while it barely responds to the initial insulin I_0 (column norm ≈ 0.08); consequently only the product-like combination is strongly determined ($\varsigma_1 \approx 73$) and the two remaining singular values are $\sim 10^3$ smaller. Thus $\mathbf{J}$ is locally full column rank 3 — the record supports *local* identifiability of (I_0, S_I, σ) — but severely ill-conditioned. Over 40 prior draws the smallest singular value has median 0.053 (IQR $[0.040, 0.11]$) and the condition number median 1.4×10^3 (IQR $[7.6 \times 10^2, 1.9 \times 10^3]$), so this is representative of the generator, not a single-point artifact. We make no claim about the continuous-time Hermann–Krener rank of the ambient six-dimensional state, which the finite five-point Jacobian (rank ≤ 5) cannot certify in any case.

This local full rank hinges on a *weak* effect. The full ISS model has hepatic glucose production depending on insulin sensitivity, HGP $=$ HGP(S_I, I) (Section 7.3). Were HGP independent of S_I, the glucose equation would be invariant under the exact one-parameter scaling

$$I \mapsto cI, \quad (N_5, N_6) \mapsto c(N_5, N_6), \quad \sigma \mapsto c\sigma, \quad S_I \mapsto S_I/c, \tag{6.8}$$

which on the restricted manifold reads $(I_0, S_I, \sigma) \mapsto (cI_0, S_I/c, c\sigma)$ (the slaved $N_{5,0}, N_{6,0}$ scale by c automatically, since $\mathrm{SS}(G_0, c\sigma) = c\,\mathrm{SS}(G_0, \sigma)$; Appendix A.4). It leaves the observed $G(t)$ unchanged, contributing an exact null direction along the log-relative tangent $\hat{t} = (+1, -1, +1)/\sqrt{3}$ and leaving only the product $S_I\sigma$ identifiable. We verify that $\hat{t}$ is indeed the operative direction rather than reading it off the singular *values*: under constant HGP the smallest right singular vector of $\mathbf{J}$ coincides with $\hat{t}$ ($|\cos| = 1.00$) at the numerical floor, so the analytic rank is ≤ 2. The full S_I-dependence of HGP breaks the symmetry only slightly: the sensitivity along $\hat{t}$ rises from $\|\mathbf{J}\hat{t}\| \approx 1 \times 10^{-4}$ (constant, numerical floor) to ≈ 0.48 (full), restoring rank 3 but leaving the product-orthogonal direction $\sim 10^3$ below the leading sensitivity. At the sparse glucose grid this weak direction is effectively unresolvable, leaving an approximate constant-$S_I\sigma$ ridge.

6.5. Sampled Local Rank versus Sparse-Sampling Conditioning. The restricted Jacobian is locally full rank at the grid, but its conditioning tells the practical story. The three singular values span $\sim 10^3$ and the product-orthogonal direction sits $\sim 10^3$ below the leading sensitivity, so it is severely ill-conditioned and therefore expected to be highly sensitive to measurement noise. Our data are noiseless, so this is a statement about *conditioning*, not a proven identifiability threshold at a specific clinical precision: the model is locally identifiable in exact arithmetic but is expected to be *practically* non-identifiable under finite precision, along the approximate constant-$S_I\sigma$ ridge.

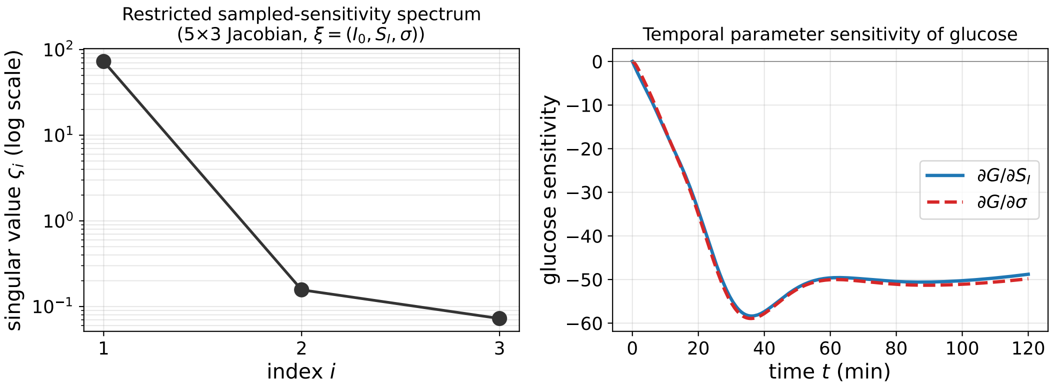


Figure 4: Restricted sampled sensitivity of the OGTT model. *(Left)* The three singular values of the 5×3 Jacobian $\nabla_\xi \mathcal{G}$, $\xi = (I_0, S_I, \sigma)$, at the five-point grid: locally full rank 3 but spanning $\sim 10^3$ (ill-conditioned). *(Right)* The temporal sensitivity of glucose to S_I and σ is nearly identical throughout the window — the two parameters act on glucose almost interchangeably, so only their stiff combination is well determined.

This ill-conditioning is temporal: the sensitivities $\partial G/\partial S_I$ and $\partial G/\partial \sigma$ (Figure 4, right) nearly coincide over the whole OGTT window, so under a uniform 30-minute grid the two parameters are almost indistinguishable from glucose. This severely limits joint recovery of (S_I, σ) and drives the regression-to-mean phenomenon along the product ridge.

The weak direction is only the HGP symmetry-breaking. To confirm that the third direction is generated *solely* by the weak S_I-dependence of hepatic glucose production, we compare the full ISS model against a *constant-HGP* ablation (HGP frozen independent of S_I and I), on the same restricted manifold, with central differences verified stable across step sizes 10^{-3}–10^{-5} (Figure 5). Three consistent signatures emerge. *(i)* The pointwise symmetry generator $I\,\partial_I \mathrm{HGP} - S_I\,\partial_{S_I}\mathrm{HGP}$ equals -1.21×10^{-2} for the full model and *exactly* 0 for constant HGP. *(ii)* Under the finite scaling $(I_0, S_I, \sigma) \mapsto (cI_0, S_I/c, c\sigma)$, the constant-HGP glucose trajectory is invariant to within the numerical integration/finite-difference tolerance $(\max_t |\Delta G|/\max_t |G| \approx 2 \times 10^{-10}$ for $c \in [1.25, 2])$, whereas the full model changes by 0.15–0.66% — the exact scaling symmetry (6.8) is present in the ablation and only weakly broken in the full model. *(iii)* Correspondingly, the sensitivity along the scaling tangent collapses from $\|\mathbf{J}\hat{t}\| \approx 0.48$ (full) to the numerical floor $\approx 1 \times 10^{-4}$ (constant), and under constant HGP the smallest right singular vector coincides with $\hat{t}$ exactly ($|\cos| = 1.00$); for the full model $\|\mathbf{J}\hat{t}\|$ *grows* with sampling density (Figure 5b). The product-orthogonal direction of the full model therefore rides entirely on a sub-percent HGP-mediated signal that a sparse, finite-precision glucose grid cannot resolve.

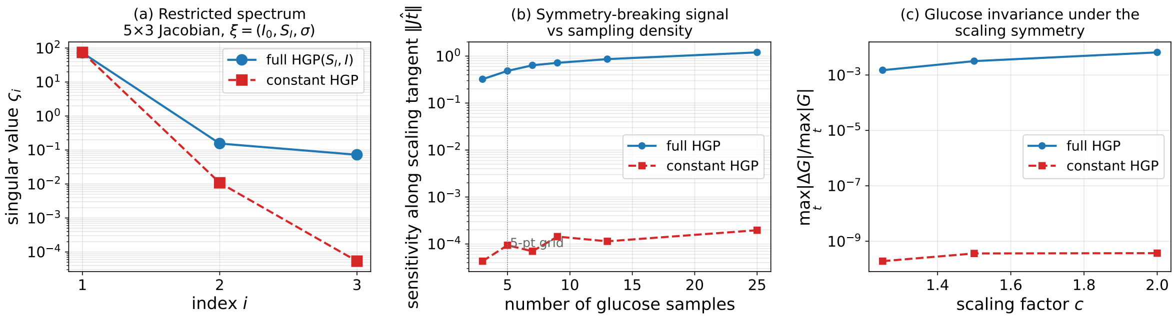


Figure 5: Full ISS-HGP versus constant-HGP ablation of the restricted OGTT sampled Jacobian $\mathbf{J} = \nabla_\xi \mathcal{G}\,\mathrm{diag}(\xi)$, $\xi = (I_0, S_I, \sigma)$ (central differences, step-size stable). *(a)* Three-value spectrum at the five-point grid: the smallest singular value drops from $\varsigma_3 \approx 0.073$ (full, numerically rank 3) to $\approx 5 \times 10^{-5}$ (constant HGP, analytic rank ≤ 2; a third value at the numerical floor). *(b)* Sensitivity $\|\mathbf{J}\hat{t}\|$ along the scaling tangent $\hat{t} = (+1, -1, +1)/\sqrt{3}$ versus number of glucose samples: for the full model it grows with density, while for constant HGP it stays at the numerical floor (exact null). *(c)* Glucose invariance under the scaling symmetry (6.8): constant-HGP glucose is invariant to within numerical tolerance (exact symmetry), whereas the full model breaks it by a sub-percent, c-dependent amount — the origin of the weak, ill-conditioned product-orthogonal direction.

Table 1: Parameter recovery (RMSE / Pearson r) on synthetic test data. The direct baseline recovers well-conditioned parameters nearly perfectly ($r \approx 0.98$); its recovery degrades along the directions the observability analysis flags as rank-deficient or ill-conditioned (LV β; OGTT S_I, σ individually). The iterative estimator converges but exhibits an additional approximation gap even where the problem is well-conditioned (SIR).

System	Parameter	Iterative (RMSE / r)	Direct baseline (RMSE / r)
SIR	β	0.107 / 0.66	**0.026 / 0.98**
(wide)	γ	0.103 / 0.69	**0.055 / 0.92**
Lotka–	α	0.102 / 0.54	**0.014 / 0.99**
Volterra	β	0.115 / 0.02	0.113 / 0.22
	δ	0.089 / 0.84	**0.012 / 0.99**
	γ	0.096 / 0.62	**0.020 / 0.99**
OGTT	S_I	0.265 / 0.65	0.212 / 0.76
	σ	0.338 / 0.65	0.254 / 0.82

7. Experiments. We evaluate the iterative fixed-point estimator on three dynamical systems that span the identifiability spectrum diagnosed a priori by the continuous- and sampled-observability analyses of Section 6: the epidemiological SIR model (locally weakly observable), the ecological Lotka–Volterra model (observability matrix rank-deficient in one parameter direction), and the physiological OGTT glucose–insulin model (weakly identifiable, with only the product $mDI = S_I\sigma$ well determined from glucose [4]). For each system we train the two networks H_ϕ, P_ψ by teacher forcing and compare the converged fixed point $\theta^*(\mathbf{x}_{\text{obs}})$ against a *direct-regression baseline* — a single network $x_{\text{obs}} \mapsto \theta$ of matched capacity trained by MSE on the identical data split. The comparison is diagnostic rather than competitive: because both estimators minimize a squared loss, both are driven toward the conditional mean (Proposition 5.5). A single finite-capacity, single-seed regressor is not the exact conditional mean, so the baseline serves as an *empirical upper benchmark* on the Bayes risk rather than an exact measurement of it; the gap between the baseline and the iteration then *suggests* an additional operator-approximation cost (Remark 5.14(ii)) on top of the shared structural term (i), without proving that the difference is purely the operator gap.

Protocol.. Synthetic trajectories are generated by numerical integration under sparse, partial observation (SIR: 2.6×10^4; LV: 2×10^4; OGTT: 5×10^4 samples). Spectral normalization is applied with a per-layer scale 0.9; for the main OGTT run this yields a certified contraction constant $\mathrm{Lip}_\mathcal{T} \leq 0.976 < 1$ (empirical Lipschitz ≈ 0.48; Section 7.4); spectral normalization enforces contractivity (Section 4.3), and accuracy is flat across margins giving $\mathrm{Lip}_\mathcal{T}$ from 0.28 to 0.92 (measured in Section 7.4); inference uses $K = 10$ fixed-point iterations. We report coordinate-wise RMSE (physical units) and Pearson correlation r between true and predicted parameters; because RMSE is scale-dependent across systems, r is the primary scale-invariant indicator of whether a parameter is recovered at all.

Table 1 summarizes all runs; the following subsections read it through the theory.

7.1. SIR: an identifiability control where the operator gap dominates. We use fixed initial conditions $(S_0, I_0, R_0) = (49, 1, 0)$ for SIR (Appendix B.1), so the only unknowns are (β, γ). The augmented SIR system satisfies the continuous-time observability rank condition (Section 6), so in the idealized limit (β, γ) are identifiable. Whether that identifiability survives the finite, sparse $S(t)$ samples actually used is a separate, empirical question, and here the answer is affirmative: the direct baseline recovers (β, γ) almost exactly ($r = 0.98, 0.92$), evidence that the conditional variance $\mathrm{Var}(\theta \mid \mathbf{x}_{\mathrm{obs}})$ is small (though a finite baseline does not prove it zero) and the structural term (Remark 5.14(i)) is negligible on this data. The iterative estimator, although it converges to a unique fixed point (Theorem 4.2), attains only $r = 0.66, 0.69$ (Figure 7): predictions concentrate in a narrow band and $\mathcal{R}_0$ saturates. Since the structural term appears small, this residual bias is plausibly dominated by the operator-approximation term (Remark 5.14(ii)) — the composition $P_\psi \circ H_\phi$ does not preserve θ perfectly, keeping the one-step residual $\varepsilon(\mathbf{x}_{\mathrm{obs}})$ bounded away from zero. SIR thus serves as a control in which the operator term dominates, and cautions against reading the iterative estimator's accuracy as evidence about identifiability. This gap is consistent with the cost of the reconstruction detour identified in Proposition 5.4: the direct regressor, which maps $\mathbf{x}_{\mathrm{obs}}$ to θ without reconstructing the hidden state, reaches the conditional-mean ceiling far more closely ($r \approx 0.98$) than the iteration ($r \approx 0.66$), on the identical data. (The same picture holds on a narrow-range variant matching commonly reported values, where the baseline reaches $r \approx 0.99$ and the iteration $r \approx 0.4$–0.5.)

To probe whether the gap is driven by the difficulty of reconstructing the hidden state — rather than by non-identifiability — we sweep the observation density (Figure 6). As the number of observation points increases from 4 to 30, the iterative estimator's recovery improves monotonically ($r_\beta : 0.66 \to 0.71 \to 0.75 \to 0.78$), consistent with denser observations making the hidden trajectory easier to reconstruct and thereby shrinking the fixed-point error through Lemma 5.2. We report this as a correlation supporting that mechanism; a direct test would additionally track the one-step residual ε, the certified q, and the bound coverage at each density, which we leave to future work. The improvement plateaus well below the direct-regressor ceiling ($r \approx 0.98$, already at 4 points), consistent with Proposition 5.4: over the tested range (≤ 30 points) the reconstruction detour retains a residual cost that added observation density does not remove. Correctness is approached, but not achieved, in the denser regime that the sparse-and-partial premise of this paper excludes.

7.2. Lotka–Volterra: observability predicts which parameters are recoverable. For prey-only observation the augmented Lotka–Volterra observability matrix is rank-deficient ($\operatorname{rank} \mathbf{O} = 5 < 6$; Section 6), the deficient direction being the joint (y_0, β) scaling symmetry (6.7). Because the hidden initial predator y_0 is a randomized nuisance in our data, marginalizing over it collapses the recovery of β specifically: the direct baseline recovers α, δ, γ nearly perfectly ($r = 0.99, 0.99, 0.99$) but collapses on β ($r = 0.22$), while every other coordinate remains recoverable (Figure 8). Thus the a-priori observability rank predicts, parameter by parameter, where marginal estimation is possible under an unknown y_0 — turning "the method sometimes fails" into a quantitative statement about the problem (had y_0 been observed, β would be identifiable). The iterative estimator recovers the same identifiable coordinates with the additional bias of (ii) ($r = 0.54$–0.84) and, like the baseline, cannot recover

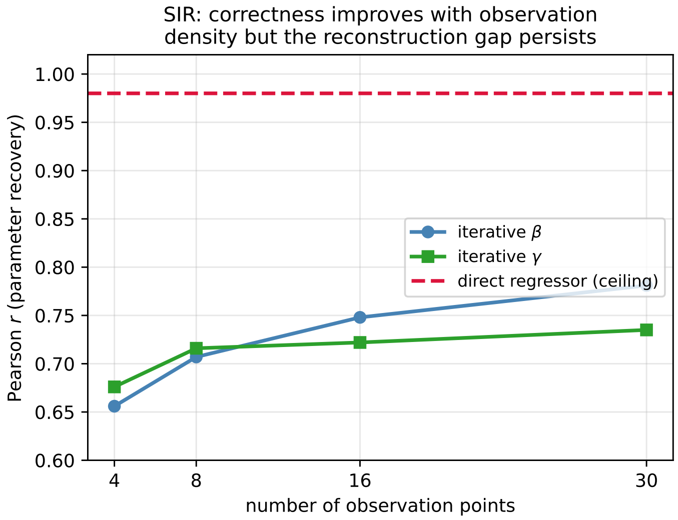


Figure 6: SIR observation-density sweep (identifiable regime, unknown-hidden-state setting; fixed initial condition). The iterative estimator's parameter recovery improves monotonically with the number of observation points — consistent with the fixed-point error being governed by the one-step residual (Lemma 5.2) — but plateaus below the direct-regression benchmark (Proposition 5.4): denser data eases hidden-state reconstruction yet does not remove the detour cost.

β from prey alone.

7.3. OGTT: collapse along an approximate product ridge and coordinate-dependent fiber selection. Data are generated by the full insulin sensitivity and secretion (ISS) model of Ha et al. [3, 4], whose glucose equation is

$$\dot{G} = \mathrm{OGTT}(t) + \mathrm{HGP}(S_I, I) - (E_{G0} + S_I I)\, G, \qquad \mathrm{HGP}(S_I, I) = \frac{h_{\max}(S_I)}{\alpha_{\mathrm{HGP}}(S_I) + I} + \mathrm{HGP}_b, \tag{7.1}$$

with $h_{\max}(S_I) = 15.443/(0.27 + S_I) - 3.54277$ and $\alpha_{\mathrm{HGP}}(S_I) = 6/(0.4 + S_I) - 0.5$; crucially, hepatic glucose production depends on S_I, which is what breaks the exact scaling symmetry (6.8) of the constant-HGP model (Section 6). Under glucose-only observation with the sparse 30-minute grid, this symmetry-breaking is too weak to recover: $\theta = (S_I, \sigma)$ is identifiable in practice only through the product $mDI = S_I\sigma$; the conditional distribution along each approximate fiber $\{S_I\sigma = c\}$ is non-degenerate, so the structural term (i) dominates and the regression-to-the-mean sandwich of Corollary 5.9 binds. One caveat is central to reading this experiment correctly: our OGTT pipeline trains MSE in *log* coordinates $u = (\log S_I, \log \sigma)$ (Appendix B.2), in which the product fiber $u_1 + u_2 = \log c$ is *affine*. By Remark 5.13 the log-coordinate Bayes rule therefore stays *on* the fiber (its back-transformed geometric mean reproduces c exactly); the arithmetic off-fiber bias of Corollary 5.12 is a property of *physical*-coordinate MSE and does *not* describe this experiment. We use the product-degeneracy

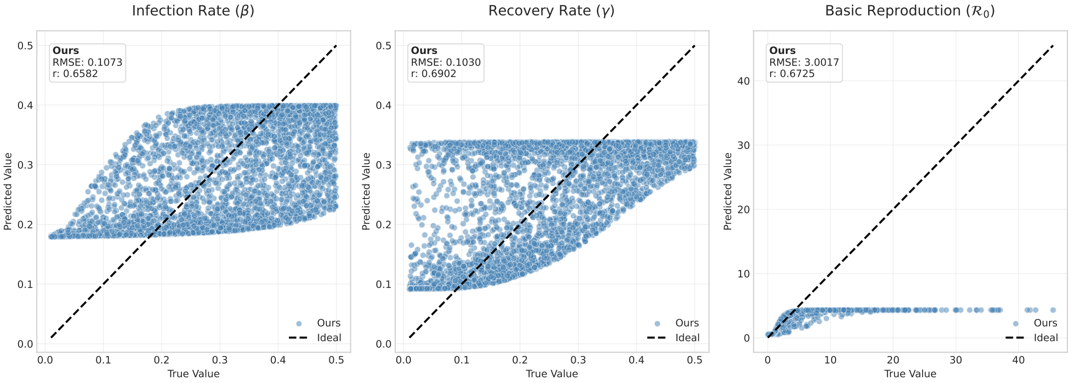


Figure 7: SIR parameter recovery, iterative estimator. Predictions (blue) versus ground truth for $\beta, \gamma, \mathcal{R}_0$. Despite guaranteed convergence, predictions compress toward a central band and $\mathcal{R}_0$ saturates — the operator-approximation gap of Remark 5.14(ii), present even though SIR is identifiable (cf. the direct baseline, $r \approx 0.98$).

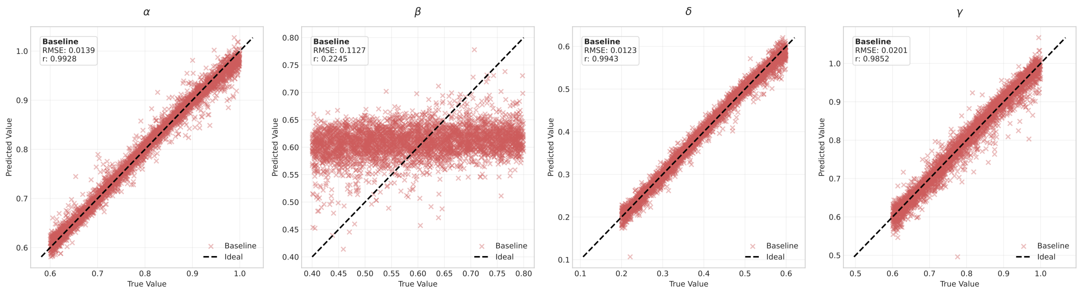


Figure 8: Lotka–Volterra recovery, direct baseline. Three parameters are recovered nearly perfectly (α, δ, γ: $r \approx 0.99$); the interaction parameter β is not ($r = 0.22$)—exactly the direction the observability matrix flags as rank-deficient (Section 6).

example only as a reference geometry, and interpret the results accordingly below.

Figure 9 makes the predicted behavior visible. *(Left)* the joint distribution of predictions collapses from the two-dimensional ground-truth cloud onto an essentially one-dimensional locus — the joint-distribution collapse of Corollary 5.9. *(Center)* the identifiable product $\log(S_I\sigma)$ (the "stiff" direction) is recovered, its density matching the ground truth (Pearson $r = 0.93$ for the iterative estimator, $r = 0.99$ for the baseline). *(Right)* the two individual coordinates are *both* weakly recovered, and to the same degree: $r(S_I) = r(\sigma) = 0.65$ for the iterative estimator and $r(S_I) = 0.76$, $r(\sigma) = 0.82$ for the baseline. Neither coordinate is privileged — only the product is pinned down, while the estimate is free to slide along the fiber. The direct baseline exhibits the *same* collapse of the individual coordinates while preserving the product, confirming that the loss of the sloppy direction binds any squared-loss point estimator and is not an artifact of the fixed-point iteration. In the log training coordinate

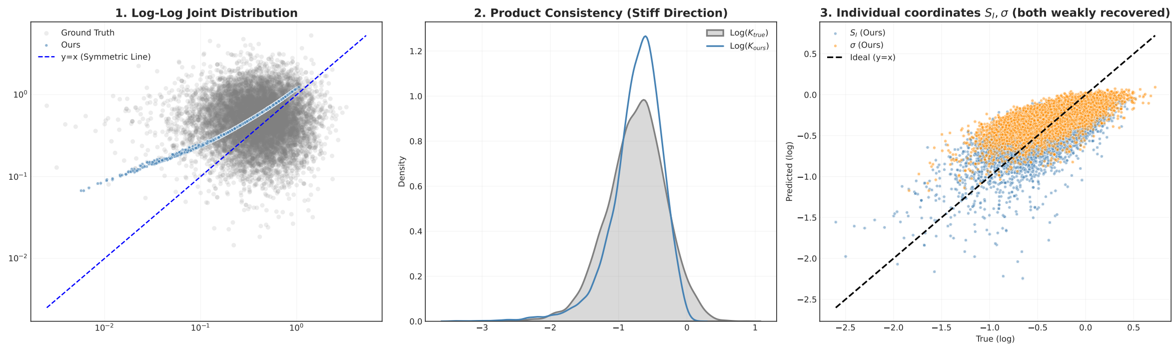


Figure 9: OGTT joint-distribution collapse (iterative estimator; the direct baseline is qualitatively identical). *Left:* predictions (blue) collapse the 2-D ground-truth cloud (grey) onto a 1-D locus. *Center:* the identifiable product $S_I\sigma$ (stiff direction) is recovered ($r = 0.93$). *Right:* the two individual coordinates S_I and σ are both weakly recovered and to the same degree ($r = 0.65$ each), neither privileged — the estimate collapses along the approximate product direction (with an additional off-fiber operator error for the iteration, quantified below). This realizes the regression-to-the-mean of Corollary 5.9 in the log training coordinate.

the product is the identifiable statistic, and both estimators recover it (the baseline nearly exactly); the iteration's larger product error is examined next and attributed to operator approximation, not to an off-fiber conditional mean.

To read the collapse geometrically we plot, as a *reference*, the product-conditioned arithmetic-mean locus obtained by binning the ground truth by $c = S_I\sigma$ and averaging within each bin, $\widetilde{m}(c) = \widehat{\mathbb{E}}[(S_I, \sigma) \mid c]$ (Figure 10, red). Two cautions are essential. First, $\widetilde{m}(c)$ is *not* the Bayes target of our estimators: it is the arithmetic conditional mean given the product, whereas the network minimizes MSE in log coordinates, whose Bayes target is the on-fiber geometric mean (Remark 5.13). That $\widetilde{m}(c)$ lies off the fibers — its product $\widetilde{m}_1\widetilde{m}_2$ exceeds c by $+0.059$ on average, in 100% of bins — is the Jensen signature of Corollary 5.12, but it is a property of the prior geometry under *physical*-coordinate averaging, not evidence that any trained estimator reached an off-fiber Bayes rule. Second, $\widetilde{m}(c)$ is conditioned on the product c, not on the observation $\mathbf{x}_{\text{obs}}$; since the full-HGP model carries weak extra information about S_I, c is not a sufficient statistic and $\widetilde{m}(c) \neq \mathbb{E}[\theta \mid \mathbf{x}_{\text{obs}}]$ in general. We therefore use $\widetilde{m}(c)$ only to visualize the fiber/prior geometry.

Read this way, the results are coherent with the log-coordinate analysis. The matched direct estimator recovers the product almost exactly (product relative error 0.9%) and sits essentially *on* the fibers (mean distance 0.004), sliding along them — precisely the on-fiber geometric-mean behavior the log-space Bayes rule predicts. The iterative fixed point, by contrast, incurs a 17% product error and sits 0.057 off the nearest fiber; because its training loss also targets the on-fiber product, this excess is attributable to *operator approximation* in the decoupled composition $P_\psi \circ H_\phi$ (Remark 5.14(ii)), not to an off-fiber conditional mean. (Its per-bin mean happens to fall nearer $\widetilde{m}(c)$ than the midpoint, but we do not read this as the network targeting the arithmetic Bayes rule, which its log-space loss does not.) The

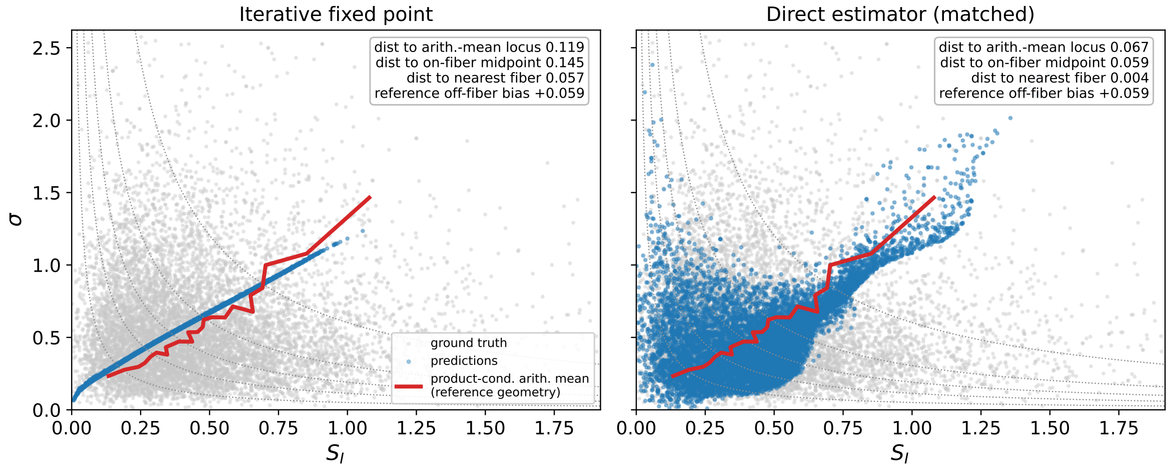


Figure 10: Predictions and the *product-conditioned arithmetic-mean locus* $\widetilde{m}(c) = \widehat{\mathbb{E}}[(S_I, \sigma) \mid S_I\sigma = c]$ in the (S_I, σ) plane (shared test split; grey: ground-truth joint, dotted: constant-$S_I\sigma$ fibers, red: $\widetilde{m}(c)$). The red curve is a *reference geometry*, not the estimators' Bayes target: the network trains in log coordinates, whose on-fiber geometric mean — not this arithmetic locus — is its target (Remark 5.13), and $\widetilde{m}(c)$ conditions on the product rather than on $\mathbf{x}_{\text{obs}}$. *Left:* the iterative fixed point collapses onto a tight 1-D band, sitting off the fibers by an amount consistent with operator approximation. *Right:* the matched direct estimator recovers the product and fans *along* the fibers (near the on-fiber geometric mean). Annotations report per-bin distances and the off-fiber product bias $\widetilde{m}_1\widetilde{m}_2 - c$ of the reference locus (positive, the Jensen property of Corollary 5.12 under physical-coordinate averaging).

reconstruction detour of Proposition 5.4 thus manifests here as degraded product consistency, consistent with a direct regressor reaching the ceiling more closely.

Physical- versus log-coordinate MSE. Remark 5.13 predicts that whether the squared-loss estimate leaves the fiber is a property of the loss coordinate, not of the degeneracy. We probe this on the generator (full HGP, where $c = S_I\sigma$ is only an approximate sufficient statistic, so this is a test in the approximate product-ridge regime rather than of the exact theorem): we train the *same* direct MSE regressor $\mathbf{x}_{\text{obs}} \mapsto (S_I, \sigma)$ under two loss coordinates — physical (S_I, σ) and logarithmic $(\log S_I, \log \sigma)$ — and measure, per test point, the product error $|\widehat{S_I\sigma} - c|/c$, the signed product bias $\widehat{S_I\sigma}/c - 1$ (positive = off the fiber, on its convex side), and the Euclidean distance to the fiber. Over five seeds (mean ± std; Figure 11):

- *log-coordinate* MSE stays *on* the fiber: product error 0.038 ± 0.004, signed bias -0.009 ± 0.011 (indistinguishable from zero), fiber distance 0.011 ± 0.002;
- *physical-coordinate* MSE leaves the fiber: product error 0.158 ± 0.038, signed bias $+0.114 \pm 0.058$ (positive — the convex side of Corollary 5.12), fiber distance 0.045 ± 0.010, four times farther off.

Crucially, the individual-coordinate recovery is *unchanged* by the coordinate ($r(S_I) = 0.78/0.77$, $r(\sigma) = 0.84/0.84$ for log/physical): the loss coordinate does not affect how well the sloppy

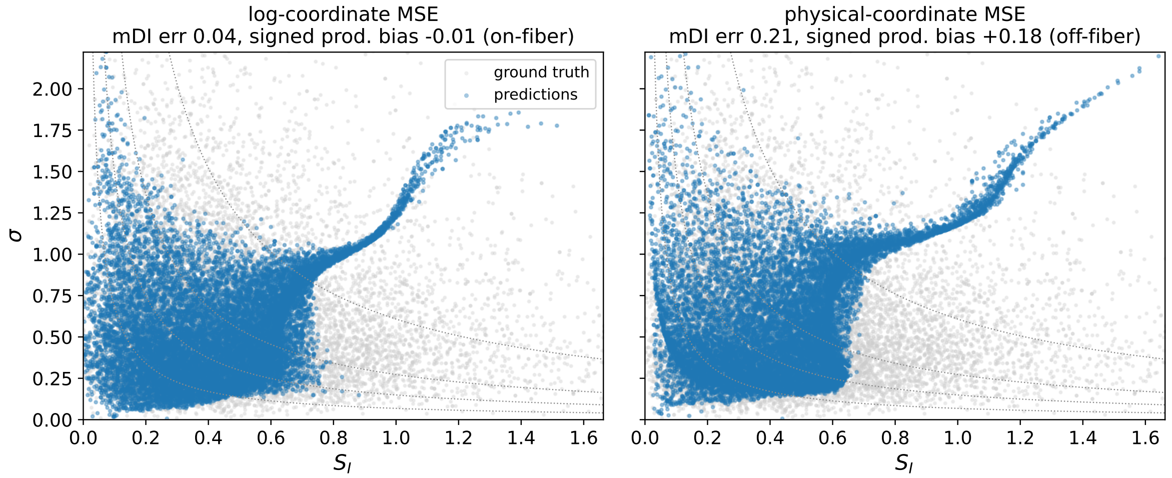


Figure 11: Physical- versus log-coordinate MSE for the direct regressor (one representative seed; grey: ground truth, dotted: constant-$S_I\sigma$ fibers, blue: predictions). Titles report the per-point product error (mDI) and signed product bias. *Left:* log-coordinate MSE keeps the product on the fiber (bias ≈ 0). *Right:* physical-coordinate MSE inflates the product (bias > 0, the Jensen/convex side of Corollary 5.12), i.e. leaves the fiber — while individual-coordinate recovery is unchanged. Off-fiber displacement is thus a property of the loss coordinate (Remark 5.13).

directions are recovered — both fail equally — only *which* representative of the ambiguous fiber the estimate prefers. This is consistent with the coordinate-dependent prediction of Remark 5.13: in this approximate product-ridge regime the off-fiber displacement of Corollary 5.12 appears under physical-coordinate MSE and is absent under the log-coordinate MSE our main pipeline uses. It also indicates, from the other side, that the iteration's product error above reflects operator approximation rather than an off-fiber conditional mean. (An exact test would repeat the ablation on the constant-HGP generator, where c is an exact sufficient statistic.)

7.4. Convergence and the role of the contraction margin. Figure 12 visualizes the fixed-point dynamics in the (S_I, σ) plane: trajectories launched from widely separated initializations all converge to a *single* fixed point (Theorem 4.2), yet that fixed point does not coincide with the ground truth — a direct picture of "convergence $\neq$ correctness."

We verify the assumptions and the bound of Lemma 5.2 over the *entire* test set ($n =$ 10,000), not a single trajectory. First, $\mathcal{T}(\cdot;\mathbf{x}_{\text{obs}})$ maps Θ into itself by construction — P_ψ terminates in a tanh, so every output lies in the normalized box $[-1,1]^p$ (the largest coordinate observed on the test set is 0.86). Second, we certify contractivity: taking the *exact* spectral norms (SVD) of the trained, spectrally normalized weights, together with the per-layer scale 0.9 and the exact activation Lipschitz constant (SiLU: 1.0998), gives a rigorous upper bound $\mathrm{Lip}_H = 0.973$, $\mathrm{Lip}_P = 1.002$, hence $\mathrm{Lip}_{\mathcal{T}} \leq \mathrm{Lip}_H \mathrm{Lip}_P = 0.976 < 1$ — a *certified* contraction (as distinct from a merely measured one). This upper bound is loose: a *diagnostic* empirical

Lipschitz quotient probed over random parameter pairs is $q_{\mathrm{emp}} = 0.48$ (a sampled lower estimate of the true constant, not a certificate). Third, the fixed-point error bound holds universally: the one-step residual has mean $\bar{\varepsilon} = 0.042$ (median 0.035) and the fixed-point error mean 0.059 (median 0.049), and every test sample satisfies $\|\theta^* - \theta^{\mathrm{true}}\| \leq \varepsilon/(1-q)$. The *rigorous* statement uses the certified $q = 0.976$, under which coverage is 100%; substituting the diagnostic $q_{\mathrm{emp}} = 0.48$ (not a certificate) tightens the bound to a still-covering median of 0.067, indicating the certificate is conservative rather than vacuous. The single annotated case of Figure 12 ($\varepsilon \approx 0.08$, $\|\theta^* - \theta^{\mathrm{true}}\| \approx 0.11$) is thus representative, not cherry-picked.

Finally, we vary the spectral-normalization margin. Across per-layer targets giving measured composite Lipschitz constants $\mathrm{Lip}_{\mathcal{T}} \in \{0.28, 0.60, 0.89, 0.92\}$, parameter accuracy is essentially flat (SIR β: RMSE 0.11–0.11, $r \approx$ 0.64–0.68); without spectral normalization the iteration *demonstrably* fails to contract — the empirical Lipschitz quotient sampled over parameter pairs exceeds one (which by itself falsifies strict contraction, unlike a mere weight-product bound) and accuracy collapses ($r < 0$). This matches the theory: spectral normalization supplies the convergence *guarantee*, but within the contractive regime the residual error is set by $\varepsilon(\mathbf{x}_{\mathrm{obs}})$ — the structural and operator terms of Remark 5.14 — not by the size of the margin.

A note on distribution shift. On real clinical OGTT measurements (out of the synthetic training distribution), the direct baseline's predictions diverge under denormalization, whereas the iterative estimator's strongly biased output remains bounded. We record this as an observation about the two estimators' behavior under distribution shift; a robustness claim is beyond the scope of the present, characterization-focused study.

8. Discussion.

It is important to locate our negative result correctly: the limitation we establish is a property of the *problem*, not of the decoupled fixed-point construction in particular. Because the conditional-mean ceiling of Proposition 5.4 applies to *any* estimator that is a function of $\mathbf{x}_{\mathrm{obs}}$ and is trained under squared loss, it binds the direct regressor and any amortized point estimator exactly as it binds our iteration. Methods built on other principles are limited too, though we state the comparison carefully. In the idealized case of an *exact, noiseless* structural degeneracy, the observation likelihood is flat along the fiber $\mathcal{F}_c$, so a likelihood-based estimator (nonlinear least squares, or EM alternating between latent-trajectory inference and parameter update) can only converge to *some* admissible fiber point — non-unique and initialization-dependent — with no indication of the residual ambiguity. Away from this idealization the picture softens: in the full OGTT model the likelihood is not flat but merely very shallow along the approximate ridge, and a constrained or fiber-projected estimator can be made to return an observation-consistent point. What no such device recovers is the *sample-specific* truth, which the observation does not determine.

We therefore avoid a clean "squared-loss off-fiber / likelihood on-fiber" dichotomy. The safe general statement is this: no deterministic point estimator can uniquely recover the sample-specific truth from an ambiguous observation, nor represent the remaining uncertainty; and whether the point it does select lies on or off the solution fiber depends on the loss, the parameterization (Remark 5.13), and any constraints or regularization. What is genuinely determined by the observation is not a single parameter but a set — the fiber, together with the distribution induced on it by the prior and any noise. The natural remedy is to

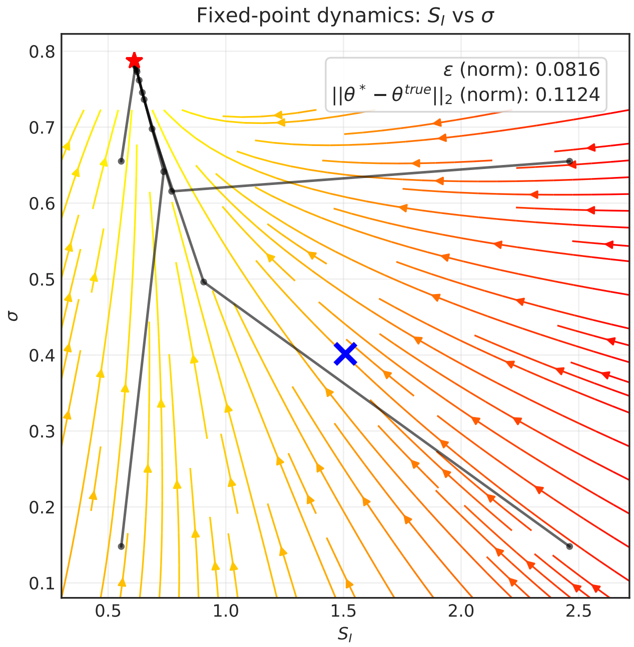


Figure 12: Fixed-point dynamics for OGTT in the (S_I, σ) plane. Trajectories from four initializations converge to one fixed point (red star), which differs from the ground truth (blue cross): convergence does not imply correctness. Annotations report the one-step residual ε and the fixed-point error, consistent with Lemma 5.2.

lift inference from point estimates to *distributions*: to characterize the posterior $p(\theta \mid \mathbf{x}_{\text{obs}})$, which represents the identifiable statistic ($mDI = S_I\sigma$ for OGTT) and the residual fiber uncertainty that point estimators cannot. Developing such a distribution-valued amortized inference procedure — one that retains the decoupled hidden-state structure but replaces the deterministic maps with conditional samplers — is the direction we pursue in subsequent work, and the ceiling identified here is precisely what it is designed to overcome.

Appendix A. Observability computations for SIR, Lotka–Volterra, and OGTT.

This appendix documents the calculations and code used to support the observability/identifiability statements in Section 6. We treat parameters as augmented constant states and follow the nonlinear observability framework of Hermann and Krener [5].

A.1. SIR model: Lie-derivative-based observability matrix. Consistently with the main text (Section 6) and with the experimental setup, we eliminate the recovered compartment through the conservation law $R = N - S - I$ and observe the *susceptible* population. The effective augmented state and observation are thus

$$z = (S, I, \beta, \gamma) \in \mathbb{R}^4, \qquad h(z) = S,$$

with the augmented vector field (the normalized-incidence SIR dynamics, parameters treated as constant states)

$$F(z) = (\dot S, \dot I, \dot\beta, \dot\gamma) = \Big(-\tfrac{\beta SI}{N},\ \tfrac{\beta SI}{N} - \gamma I,\ 0,\ 0\Big).$$

The first Lie derivatives of $h = S$ along F and their gradients are

$$L_F^0 h(z) = S, \qquad \nabla_z L_F^0 h = (1,0,0,0),$$
$$L_F^1 h(z) = -\tfrac{\beta SI}{N}, \qquad \nabla_z L_F^1 h = \Big(-\tfrac{\beta I}{N},\ -\tfrac{\beta S}{N},\ -\tfrac{SI}{N},\ 0\Big),$$

and continuing through $L_F^2 h,\ L_F^3 h$ we assemble the observation map and the 4×4 observability matrix

$$\Phi(z) = \big[\, L_F^0 h,\ L_F^1 h,\ L_F^2 h,\ L_F^3 h \,\big]^\top, \qquad \mathbf{O}(z) := \nabla_z \Phi(z) \in \mathbb{R}^{4\times 4}.$$

The following `sympy` script constructs $\mathbf{O}(z)$ and evaluates its determinant symbolically:

```
import sympy as sp
S, I, beta, gamma, N = sp.symbols('S I beta gamma N', positive=True)
z = sp.Matrix([S, I, beta, gamma])
F = sp.Matrix([-beta*S*I/N, beta*S*I/N - gamma*I, 0, 0])
def Lie(expr):
    return (sp.Matrix([expr]).jacobian(z) * F)[0]
h = S
rows, cur = [h], h
for _ in range(3):
    cur = sp.expand(Lie(cur)); rows.append(cur)
O = sp.Matrix([list(sp.Matrix([r]).jacobian(z)) for r in rows])
print("det O_S =", sp.factor(sp.simplify(O.det())))   # I**3*S**4*beta**4/N**5
print("rank    =", O.rank())                           # 4
```

Running it yields the closed-form determinant

$$\det \mathbf{O}_S(z) \;=\; \frac{I^3 S^4 \beta^4}{N^5},$$

which is nonzero for every generic state with $S, I, \beta \neq 0$; hence $\operatorname{rank} \mathbf{O}_S(z) = 4$ and the SIR system is locally weakly observable (locally identifiable at generic states) from $S(t)$ alone. For reference, the same construction with the alternative observation $h(z) = I$ gives

$$\det \mathbf{O}_I(z) \;=\; -\frac{I^6 S \beta^4}{N^5},$$

so the conclusion (rank 4, generic observability) is identical although the algebra differs; the main text and experiments use the S-only observation. We stress that this is a *continuous-time* rank condition on exact outputs and their derivatives; its relation to the finite, sparsely sampled observations actually available is taken up in Section 6.

A.2. Lotka–Volterra model: symbolic rank deficiency. For the Lotka–Volterra system we augment the state as

$$z = (x, y, \alpha, \beta, \gamma, \delta),$$

with vector field

$$\Sigma(z) = (\alpha x - \beta xy,\ \delta xy - \gamma y,\ 0,\ 0,\ 0,\ 0),$$

and observation $h(z) = x$.

The following `sympy` script computes the observation matrix and its rank:

```
import sympy as sp

x, y, a, b, g, d = sp.symbols('x y a b g d')
z = sp.Matrix([x, y, a, b, g, d])

sigma = sp.Matrix([
    a*x - b*x*y,
    d*x*y - g*y,
    0, 0, 0, 0
])

h = x

Phi = []
current_L = h
Phi.append(current_L)

for _ in range(len(z) - 1):     # adds L^1..L^5 -> Phi = L^0..L^5 (6 rows)
    grad_L = sp.Matrix([current_L]).jacobian(z)
    current_L = (grad_L * sigma)[0]
    Phi.append(current_L)

O = sp.Matrix(Phi).jacobian(z)  # 6 x 6 observation matrix
print("rank(O) =", O.rank())
```

Running this code yields `rank(O) = 5`, while the augmented state dimension is 6. Thus the observation matrix is rank-deficient and the Lotka–Volterra model is not locally observable from prey-only observations. Figure 3 in the main text provides a concrete example of two distinct states/parameters producing essentially identical $x(t)$.

A.3. OGTT model: restricted sampled-sensitivity rank and conditioning. For OGTT we analyze the sampled sensitivity on the free-variable manifold $\xi = (I_0, S_I, \sigma)$ (Section 6.4): $G_0 = G(t_0)$ is observed, and the incretin initial states are slaved to $(N_{5,0}, N_{6,0}) = \mathrm{SS}(G_0, \sigma)$, so σ-perturbations recompute the steady state. We use *central* differences with a combined relative-plus-absolute step $h_k = \texttt{atol} + \texttt{rtol} \cdot |\xi_k|$; a purely relative step $\mathrm{d}\xi_k = \xi_k \varepsilon$ would degenerate at $\xi_k = 0$. Columns are normalized by the ξ-scales so the singular-value comparison is dimensionless.

```
# xi = (I0, S_I, sigma); G0 fixed; (N5,N6)=SS(G0,sigma) recomputed in simulate_G
def restricted_jacobian(simulate_G, xi, times, rtol=1e-4, atol=1e-6):
    J = np.zeros((len(times), 3)); cs = np.abs(xi)
    for k in range(3):
        h = atol + rtol * abs(xi[k])       # never zero
        ep = xi.copy(); ep[k] += h
        em = xi.copy(); em[k] -= h
        dG = simulate_G(ep, times) - simulate_G(em, times)
        J[:, k] = dG / (2*h) * cs[k]
    S = np.linalg.svd(J, compute_uv=False)  # 5x3 Jacobian: <= 3 svals
    return S, S[0] / S[-1]                   # sing. values, cond. number
```

Here $\texttt{col\_scale}_k = |\xi_k|$, so J is the *log-relative* Jacobian $D_\xi \mathcal{G}\ \mathrm{diag}(\xi)$ (columns $\partial G/\partial \log \xi_k$; glucose in raw mg/dL); this is a scale-invariant relative-coordinate diagnostic, distinct from the network training coordinate (Appendix B.2). We evaluate at the nominal point $G_0 = 91$ mg/dL, $\xi = (I_0, S_I, \sigma) = (6, 0.5, 0.5)$, check stability across $\texttt{rtol} \in [10^{-5}, 10^{-3}]$, and use rank tolerance $10^{-3}\varsigma_{\max}$. At the five-point grid J is 5×3 (its first row is zero since $G(t_0) = G_0$ is fixed) and yields $\varsigma \approx (73, 0.16, 0.073)$ (condition number $\approx 10^3$): locally full column rank 3 but severely ill-conditioned. Under the constant-HGP ablation the third singular value drops to $\approx 5 \times 10^{-5}$ and its right singular vector coincides exactly with the scaling tangent $\hat{t} = (+1, -1, +1)/\sqrt{3}$ ($|\cos| = 1.00$), so the analytic rank is ≤ 2. We make no claim about the continuous-time Hermann–Krener rank of the ambient six-dimensional state; the finite five-point Jacobian (rank ≤ 5) cannot certify it.

A.4. Homogeneity behind the constant-HGP scaling symmetry. We justify the exact symmetry $(I_0, S_I, \sigma) \mapsto (cI_0, S_I/c, c\sigma)$ used in Section 6.4 for the constant-HGP model. Write the ISS insulin/incretin subsystem schematically as

$$\dot{I} = \tfrac{b}{\mathrm{BV}}\,\mathrm{ISR}(N_5) - kI, \qquad \dot{N}_5 = a_{55}N_5 + a_{56}N_6, \qquad \dot{N}_6 = \sigma\, r(G) + a_{65}N_5 + a_{66}N_6,$$

where the coefficients a_{ij} and $r(G)$ depend on G but not on σ or S_I, and ISR is *linear* in N_5 (a product of G-dependent pool fractions times N_5; see `systems/ogtt_simul.py`). The secretion parameter σ enters only as a multiplicative gain on the incretin forcing $\sigma r(G)$. Consequently the map $(I, N_5, N_6) \mapsto c\,(I, N_5, N_6)$ together with $\sigma \mapsto c\sigma$ leaves these three equations invariant (each side scales by c), and the slaved steady state obeys $\mathrm{SS}(G_0, c\sigma) = c\,\mathrm{SS}(G_0, \sigma)$ because it solves a linear system whose forcing is $\propto \sigma$. Glucose couples to insulin only through the term $-(E_{G0} + S_I I)G$; under $I \mapsto cI$ and $S_I \mapsto S_I/c$ the product $S_I I$ is invariant, and with constant HGP the entire glucose equation is unchanged. Since $G(0) = G_0$ is held fixed, $G(t)$ is exactly invariant. (For the full ISS model, HGP depends on S_I, so this invariance is only approximate — the weak breaking quantified in Section 6.4.)

Appendix B. Reproducibility: systems, data, architecture, and training.

This appendix records the systems, data-generating distributions, network architectures, and training protocol used in Section 7. The SIR and Lotka–Volterra models are given in full below; the OGTT ISS model has a large secretion/incretin subsystem that we do not reproduce in its entirety here, deferring its full equations and fixed physiological constants to the source implementation [3] and the released code. Exact reproduction of every reported number therefore requires the accompanying code at the fixed commit (Appendix B.5); the values below are taken from that configuration.

B.1. Benchmark systems and governing equations. Each system is integrated as an initial value problem with `scipy.integrate.solve_ivp` at the listed sampling times; a single state variable is observed and the rest is treated as hidden. Parameters are treated as constant augmented states for the observability analysis of Section 6.

SIR (epidemiological). With $N = S + I + R$ conserved,

$$\dot{S} = -\frac{\beta S I}{N}, \qquad \dot{I} = \frac{\beta S I}{N} - \gamma I, \qquad \dot{R} = \gamma I,$$

observed variable S, hidden variable I, parameters $\theta = (\beta, \gamma)$. Fixed initial condition $(S_0, I_0, R_0) = (49, 1, 0)$. Priors: β, γ uniform, on the *wide* range $\beta, \gamma \sim \mathcal{U}[0.01, 0.5]$ (a *narrow* variant uses $\beta \sim \mathcal{U}[0.08, 0.12]$, $\gamma \sim \mathcal{U}[0.09, 0.11]$). The ODE is integrated on $[0, 110]$ and observations are taken on the uniform grid $t = \text{linspace}(0, 100, N_{\text{obs}})$ with N_{obs} points ($N_{\text{obs}} = 8$ for the main runs; the density sweep uses $N_{\text{obs}} \in \{4, 8, 16, 30\}$); the extra interval past the last observation is a small integration buffer.

Lotka–Volterra (ecological).

$$\dot{x} = \alpha x - \beta x y, \qquad \dot{y} = \delta x y - \gamma y,$$

observed prey x, hidden predator y, parameters $\theta = (\alpha, \beta, \delta, \gamma)$. Priors $\alpha \sim \mathcal{U}[0.6, 1.0]$, $\beta \sim \mathcal{U}[0.4, 0.8]$, $\delta \sim \mathcal{U}[0.2, 0.6]$, $\gamma \sim \mathcal{U}[0.6, 1.0]$; initial conditions $x_0 \in [5, 15]$, $y_0 \in [1, 5]$ accepted by rejection so that the conserved-energy offset $\Delta V = V(x_0, y_0) - V(x_{\text{eq}}, y_{\text{eq}})$ lies in $[0.1, 1.5]$ (ensuring non-trivial oscillations), where $V(x, y) = \delta x - \gamma \log x + \beta y - \alpha \log y$ is the standard Lotka–Volterra conserved quantity and $(x_{\text{eq}}, y_{\text{eq}}) = (\gamma/\delta, \alpha/\beta)$. Time span $[0, 20]$, 10 uniform samples $t = \text{linspace}(0, 20, 10)$.

OGTT (physiological, full ISS model). The glucose equation is (7.1), coupled to insulin and a two-stage incretin delay chain,

$$\dot{I} = \frac{b\,\text{ISR}(N_5; S_I, \sigma)}{\text{BV}} - k\,I, \qquad \dot{N}_5,\ \dot{N}_6 : \ \text{incretin delay dynamics},$$

with hepatic glucose production $\text{HGP}(S_I, I)$ as in Section 7.3. Observed glucose G, hidden insulin I, parameters $\theta = (S_I, \sigma)$. Parameters and the initial glucose/insulin are drawn from data-driven log-normal priors (`scipy.stats.lognorm`, rejection-sampled for strict positivity), and $(N_{5,0}, N_{6,0})$ are set to the algebraic steady state given G_0: writing $\text{LogN}(s, \text{loc}, \text{scale})$ for the `scipy.stats.lognorm` parameters,

$$S_I \sim \text{LogN}(0.527, -0.105, 0.506), \quad \sigma \sim \text{LogN}(0.593, -0.053, 0.543),$$

$$G_0 \sim \text{LogN}(0.286, 60.07, 29.81), \quad I_0 \sim \text{LogN}(0.596, -0.013, 6.495).$$

Time span $[0, 120]$; the sparse clinical 30-minute grid $t \in \{0, 30, 60, 90, 120\}$ (5 points). The fixed physiological constants (E_{G0}, b, k, BV, the HGP and incretin coefficients) are those of the released ISS implementation [3].

Observations. All training/test data in Section 7 are *noiseless* deterministic ODE integrations (the stochastic-forcing option and the derivative-feature option are disabled). The difficulty comes from the ambiguity and ill-conditioning induced by partial, sparse sampling — not from added measurement noise; SIR is an identifiable control, whereas Lotka–Volterra and the full OGTT model are, respectively, unidentifiable and only weakly/locally identifiable from partial observations.

B.2. Data splits and normalization. Sample sizes: OGTT 5×10^4, SIR 2.6×10^4, Lotka–Volterra 2×10^4 trajectories. Each set is partitioned 60/20/20 into train/validation/test with a fixed seed, the split performed *before* normalization so that the normalizer is fit on the training set only (no leakage). For the SIR generalization discussion we additionally use a parameter-*extrapolation* (out-of-distribution) test split; unless noted, results use the standard random split.

Normalization: observed and hidden states are divided by a per-variable scale (the training-set maximum absolute value). Parameters are mapped to $[-1, 1]$ by a min–max transform fit on the training set with a 10% margin; for OGTT this is applied after a log transform (so the multiplicative $S_I\sigma$ structure becomes additive), while SIR and LV use the plain min–max. Denormalization inverts these maps. The fixed-point iteration is initialized at $\theta^{(0)}$, the geometric mean of the training parameters.

B.3. Network architecture and matched-capacity baseline. Both H_ϕ and P_ψ are multilayer perceptrons with hidden widths $[256, 256, 256]$ and SiLU activations. When spectral normalization is enabled, every linear layer is spectrally normalized (one power iteration) and each hidden block is rescaled by a fixed per-layer factor 0.9 (the `ExcludeLambda` module); P_ψ additionally ends in a tanh, which bounds its output to the normalized box $[-1, 1]^p$ and thereby guarantees $\mathcal{T}(\cdot; \mathbf{x}_{\text{obs}})$ maps Θ into Θ (Section 7.4). Weights use orthogonal initialization. Parameter counts (spectral-normalized): for OGTT, $|\phi| = 134{,}917$ and $|\psi| = 134{,}914$ ($\approx 2.70 \times 10^5$ total); SIR $\approx 2.69 \times 10^5$; Lotka–Volterra $\approx 2.76 \times 10^5$.

The *matched-capacity direct baseline* (`SingleNetworkBaseline`) maps $\mathbf{x}_{\text{obs}} \mapsto \theta$ through the *same* MLP backbone — identical hidden widths $[256, 256, 256]$, SiLU activations, spectral normalization, initialization, optimizer, and data split — as a single network (≈ 1.33–1.35×10^5 parameters, i.e. one branch of the operator). "Matched capacity" thus means an identical per-network architecture and training recipe, isolating the effect of the decoupled reconstruction structure rather than of raw parameter count.

B.4. Training and inference hyperparameters. The main recovery numbers (Table 1) and the trained-operator figures come from single-seed runs (seed 42) of the full training pipeline; retraining that pipeline across many seeds for all three systems is GPU-bound and, due to computational constraints, was not carried out here, so we report point values there. Where multi-seed evidence is inexpensive we do provide it: the coordinate-MSE ablation of Section 7.3 is run over five seeds and reported as mean $\pm$ std, and it confirms that the qualitative conclusions (the on/off-fiber behavior and the individual-coordinate recovery) are seed-stable. The certified contraction constant and the test-set-wide residual/error statistics of Section 7.4 are computed from the saved checkpoint of the main OGTT run.

Table 2: Training and inference configuration (identical across the three systems unless a value is listed per system).

Setting	Value
Optimizer	Adam, learning rate 10^{-3}, weight decay 0
LR schedule	`ReduceLROnPlateau` (factor 0.9, patience 200, min 10^{-8})
Batch size	256
Epochs (max)	1000 (SIR, LV); 1500 (OGTT)
Early stopping	on validation loss, patience 150 (SIR, LV) / 250 (OGTT), min-delta 10^{-6}
Training losses	decoupled teacher forcing: $\mathcal{L}_H + \mathcal{L}_P$ (both MSE), Eqs. (3.6)–(3.7)
Spectral-norm scale	0.9 per hidden layer (contraction runs)
Inference iterations	$K = 10$ fixed-point steps from $\theta^{(0)}$
Random seed	42 (data generation, split, and torch)

B.5. Code and scripts. The code, configurations, and analysis scripts will be released publicly upon publication, at a fixed commit that reproduces every reported number. The experiments of this pass are reproduced by standalone scripts under `experiments/det_meanlocus/`: `condmean_locus_fig.py` (conditional-mean locus, Figure 10), `ogtt_observability.py` (full- vs constant-HGP sampled sensitivity, Figure 5), `certify_contraction.py` (contraction certification), and `coord_mse_ablation.py` (physical- vs log-coordinate MSE ablation, Figure 11). Model training uses `run_paper_exp.py` with the settings of Table 2.

REFERENCES

[1] S. Bai, J. Z. Kolter, and V. Koltun, *Deep equilibrium models*, in Advances in Neural Information Processing Systems (NeurIPS), 2019.

[2] K. Cranmer, J. Brehmer, and G. Louppe, *The frontier of simulation-based inference*, Proceedings of the National Academy of Sciences, 117 (2020), pp. 30055–30062.

[3] J. Ha, S. T. Chung, M. Springer, J. Y. Kim, P. Chen, A. Chhabra, M. G. Cree, C. Diniz Behn, A. E. Sumner, S. A. Arslanian, and A. S. Sherman, *Estimating insulin sensitivity and β-cell function from the oral glucose tolerance test: validation of a new insulin sensitivity and secretion (ISS) model*, American Journal of Physiology-Endocrinology and Metabolism, 326 (2024), pp. E454–E471, https://doi.org/10.1152/ajpendo.00189.2023.

[4] J. Ha, J. Y. Kim, M. Springer, A. Chhabra, S. T. Chung, A. E. Sumner, A. S. Sherman, and S. Arslanian, *A mathematical model-derived disposition index without insulin validated in youth with obesity*, Journal of Clinical Endocrinology & Metabolism, 110 (2025), pp. 706–714, https://doi.org/10.1210/clinem/dgae582.

[5] R. Hermann and A. Krener, *Nonlinear controllability and observability*, IEEE Transactions on Automatic Control, 22 (1977), pp. 728–740.

[6] W. O. Kermack and A. G. McKendrick, *A contribution to the mathematical theory of epidemics*, Proceedings of the royal society of london. Series A, Containing papers of a mathematical and physical character, 115 (1927), pp. 700–721.

[7] A. J. Lotka, *Elements of physical biology*, Williams & Wilkins, 1925.

[8] T. Miyato, T. Kataoka, M. Koyama, and Y. Yoshida, *Spectral normalization for generative adversarial networks*, 2018, https://arxiv.org/abs/1802.05957.

[9] G. Papamakarios and I. Murray, *Fast ε-free inference of simulation models with Bayesian conditional density estimation*, in Advances in Neural Information Processing Systems (NeurIPS), 2016.

[10] R. R. Phelps, *Lectures on Choquet's Theorem*, vol. 1757 of Lecture Notes in Mathematics, Springer,

2 ed., 2001.

[11] F. Takens, *Detecting strange attractors in turbulence*, in Dynamical Systems and Turbulence, Warwick 1980, vol. 898 of Lecture Notes in Mathematics, Springer, 1981, pp. 366–381.

[12] A. Tarantola, *Inverse problem theory and methods for model parameter estimation*, SIAM, 2005.

[13] A. F. Villaverde, A. Barreiro, and A. Papachristodoulou, *Structural identifiability of dynamic systems biology models*, PLoS Computational Biology, 12 (2016), p. e1005153.

[14] V. Volterra, *Fluctuations in the abundance of a species considered mathematically*, Nature, 119 (1927), pp. 12–13.

[15] R. J. Williams and D. Zipser, *A learning algorithm for continually running fully recurrent neural networks*, Neural computation, 1 (1989), pp. 270–280.